



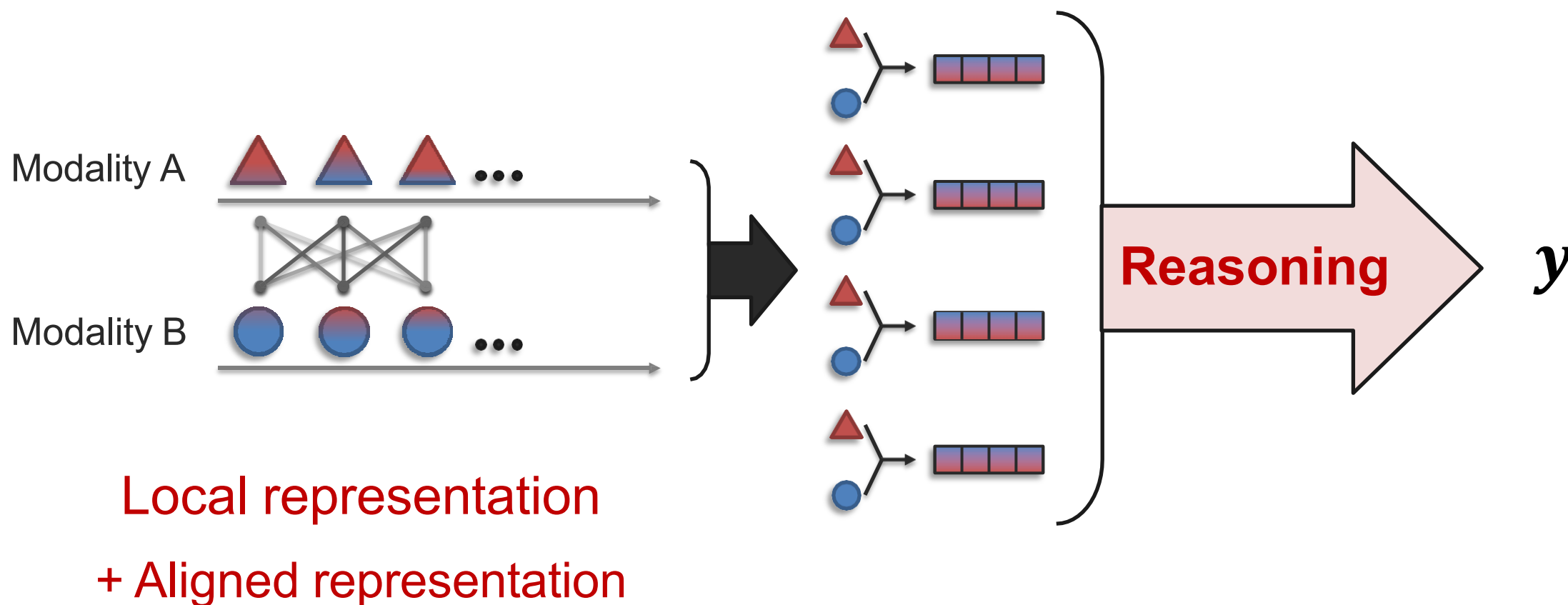
Multimodal Machine Learning

Lecture 7.2: Multimodal Inference and Knowledge

Dam Quang Tuan

Reasoning

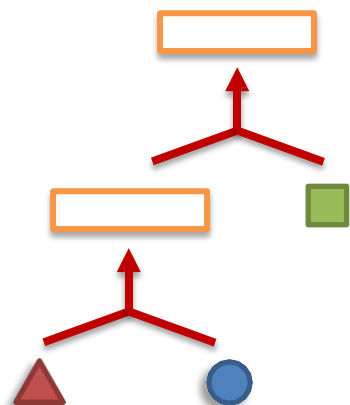
Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.



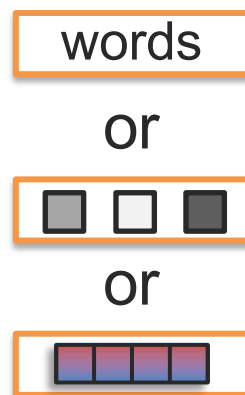
Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.

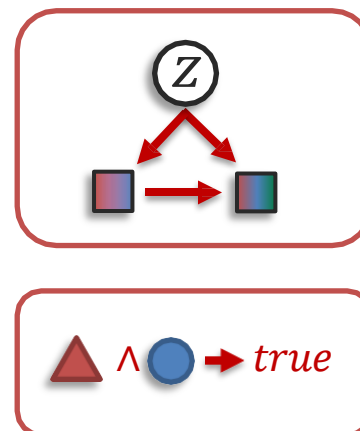
(A) Structure modeling



(B) Intermediate concepts



(C) Inference paradigm



(D) External knowledge



Summary

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.

(A) Structure modeling

Temporal
Hierarchical

Interactive

(B) Intermediate concepts

Continuous

(C) Inference paradigm

Causal
Logical

(D) External knowledge

Knowledge
Commonsense

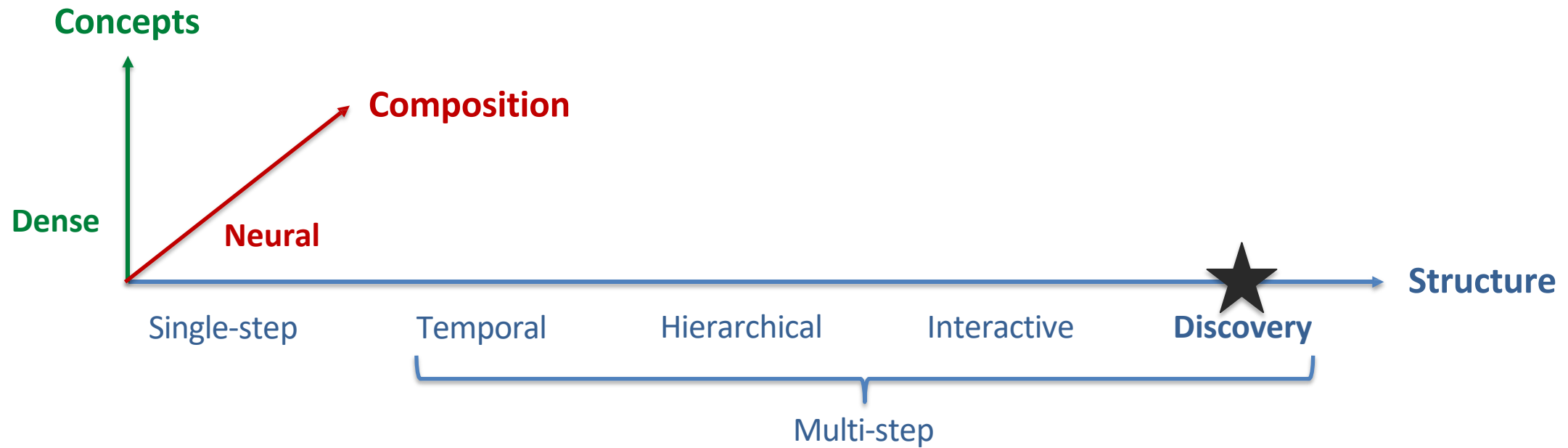
Today

Discovery

Discrete



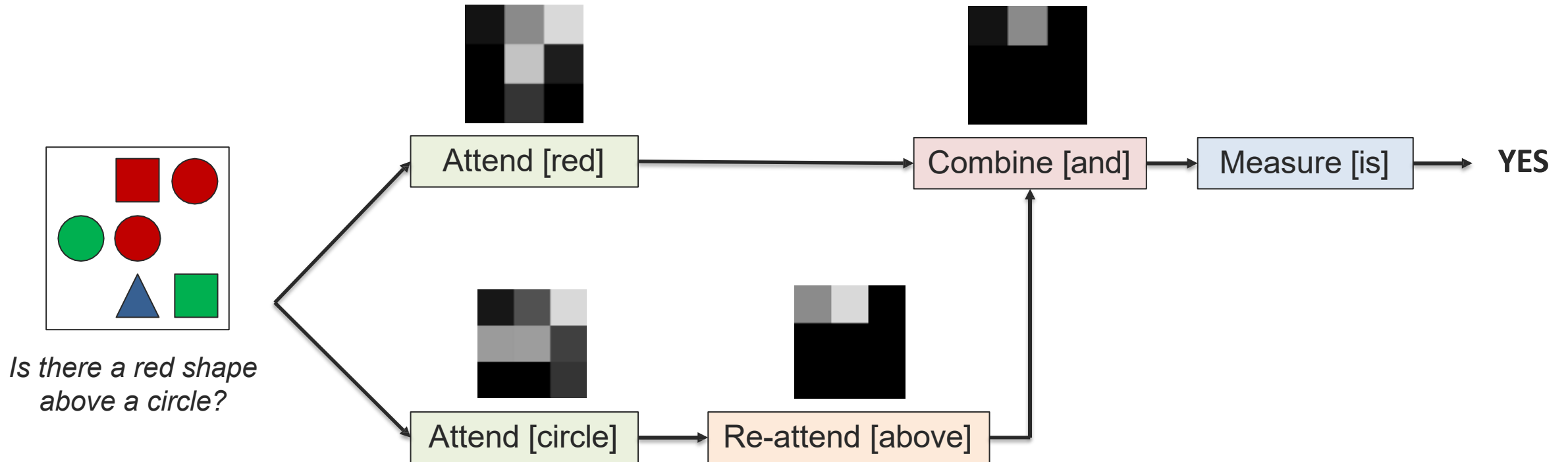
Sub-Challenge 3a: Structure Modeling



Structure Discovery

End-to-end neural module networks

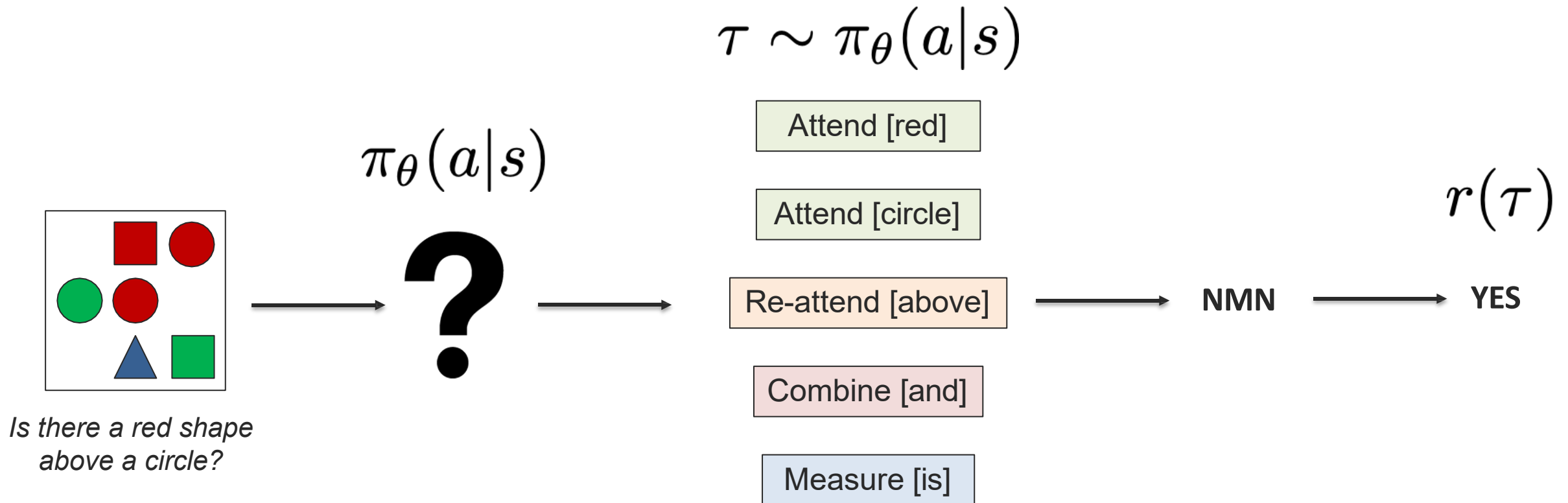
Recall structure - leverage syntactic structure of language based on parsing



Structure Discovery

End-to-end neural module networks

Can we learn the structure end-to-end?



Stochastic Optimization

$$\max_{\theta} \mathbb{E}_{q_{\theta}(\mathbf{z})} [f(\mathbf{z})]$$

RL

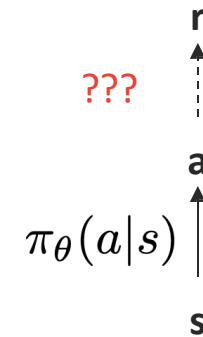
$$\max_{\theta} J(\theta) \quad \text{Reward}$$

$$\max_{\theta} \mathbb{E}_{\tau \sim p(\tau; \theta)} [r(\tau)]$$

In RL (at least for discrete actions):

- T is a sequence of discrete actions
- $p(T; \theta)$ is not reparameterizable
- $r(T)$ is a black box function
i.e. the environment

???



REINFORCE is a general-purpose solution!

Revisiting REINFORCE

$$\max_{\theta} \mathbb{E}_{q_{\theta}(\mathbf{z})} [f(\mathbf{z})] \quad (\text{we will revisit this equation for generative models})$$

We want to take gradients wrt θ of the term:

$$\nabla_{\theta} \mathbb{E}_{q_{\theta}(\mathbf{z})} [f(\mathbf{z})] = \mathbb{E}_{q_{\theta}(\mathbf{z})} [f(\mathbf{z}) \nabla_{\theta} \log q_{\theta}(\mathbf{z})]$$

We can now compute a Monte Carlo estimate:

Sample $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^K$ from $q_{\theta}(\mathbf{z})$ and estimate

$$\nabla_{\theta} \mathbb{E}_{q_{\theta}(\mathbf{z})} [f(\mathbf{z})] \approx \frac{1}{K} \sum_k [f(\mathbf{z}^k) \nabla_{\theta} \log q_{\theta}(\mathbf{z}^k)]$$

What we derived: sample trajectories and compute:

$$\nabla_{\theta} J(\theta) \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

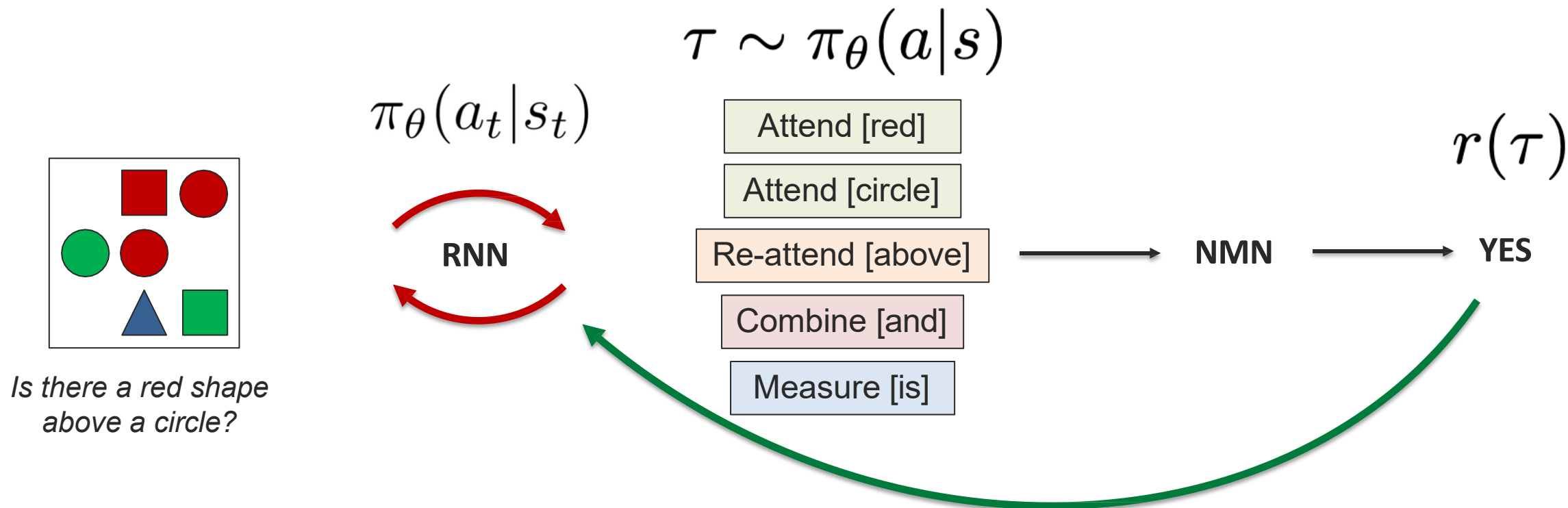
- $\mathbf{z}$ can be discrete or continuous!
- $q(\mathbf{z})$ can be a discrete and continuous distribution!
- $q(\mathbf{z})$ must allow for easy sampling and be differentiable wrt θ
- $f(\mathbf{z})$ can be a black box!

Structure Discovery

End-to-end neural module networks

Can we learn the structure end-to-end?

$$\nabla_{\theta} J(\theta) \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$



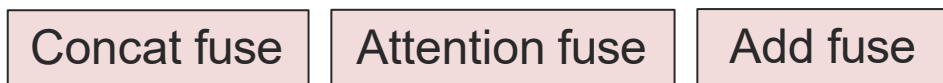
Structure Discovery

Structure fully learned from optimization and data

1. Define basic representation building blocks



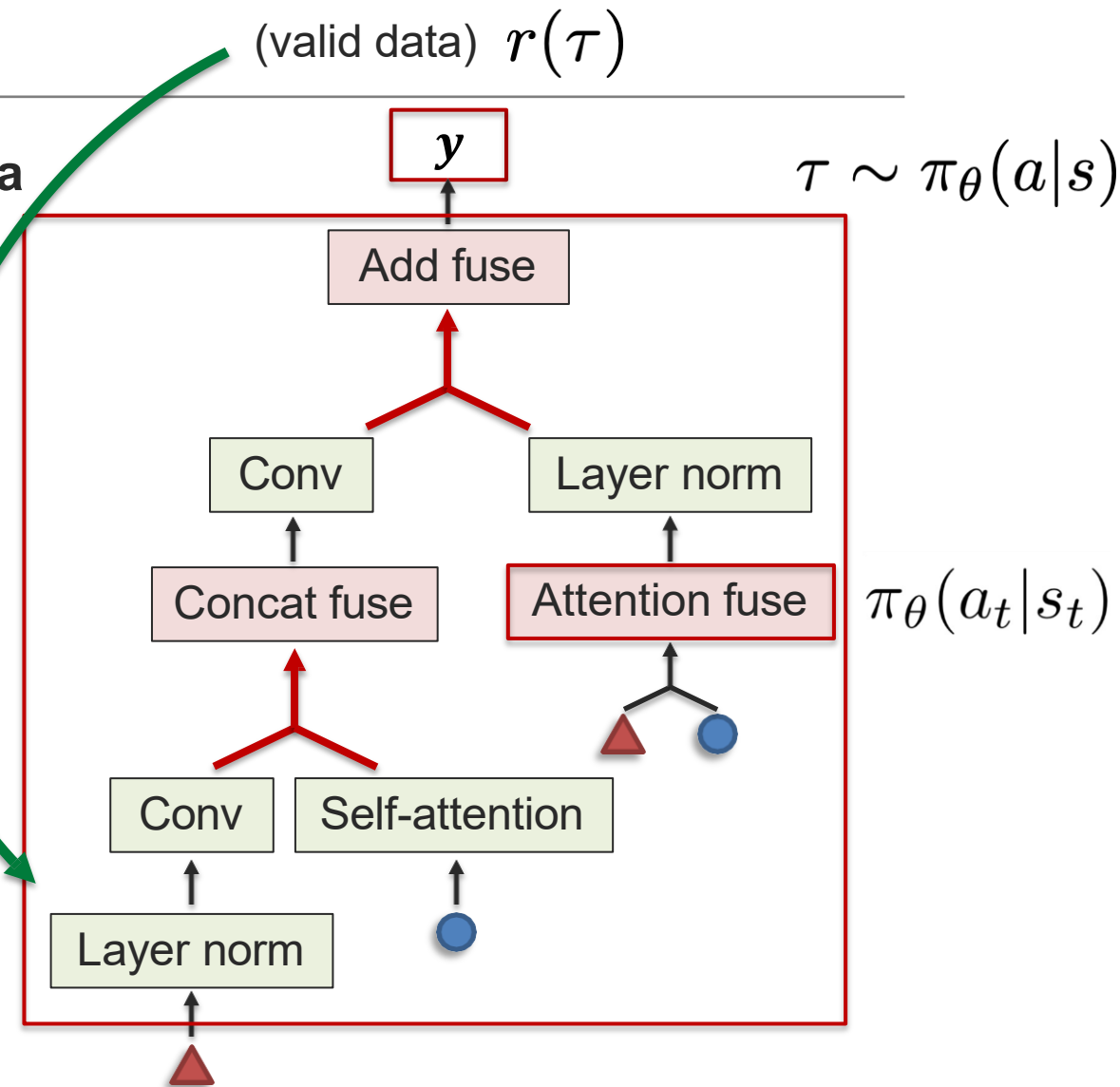
2. Define basic fusion building blocks



3. Automatically search for composition using neural architecture search

$$\nabla_{\theta} J(\theta) \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

Nice, but slow!



Continuous Structure Discovery

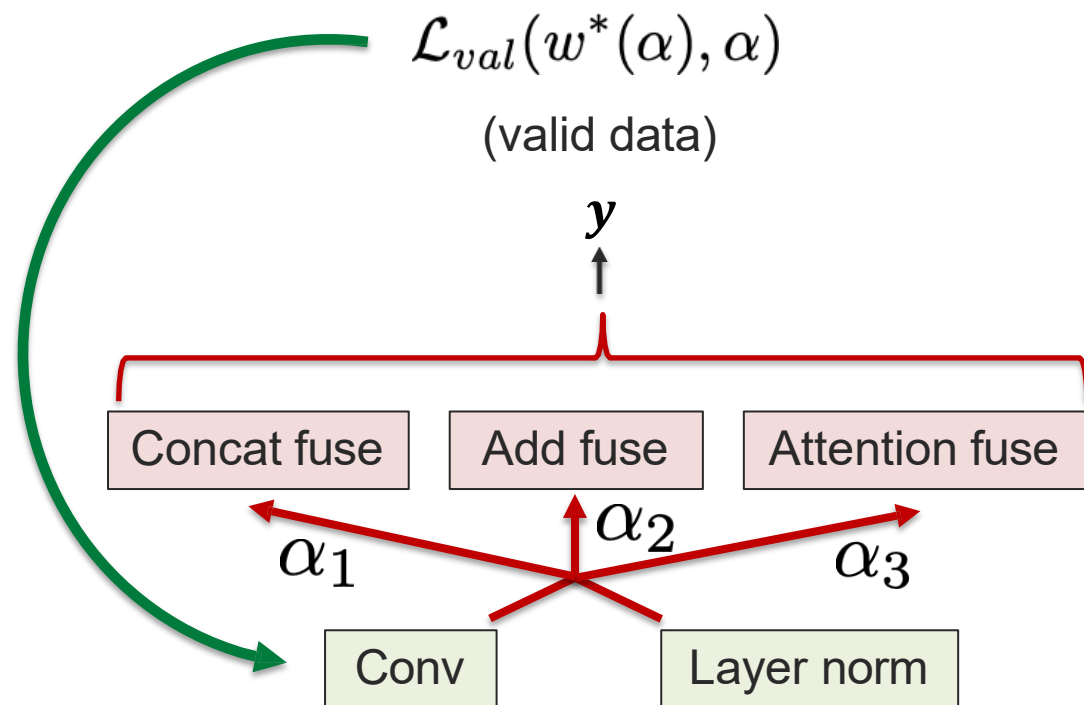
Biggest problem: discrete optimization is slow.
Differentiable optimization for structure learning:

1. Approximate selection with softmax:

$$o'(x) = \sum_i \frac{\exp(\alpha_i)}{\sum_i \exp(\alpha_i)} o_i(x)$$

2. Solve bi-level optimization problem

$$\begin{aligned} \min_{\alpha} \quad & \mathcal{L}_{val}(w^*(\alpha), \alpha) \\ \text{s.t.} \quad & w^*(\alpha) = \operatorname{argmin}_w \mathcal{L}_{train}(w, \alpha) \end{aligned}$$



Continuous Structure Discovery

Biggest problem: discrete optimization is slow.
Differentiable optimization for structure learning:

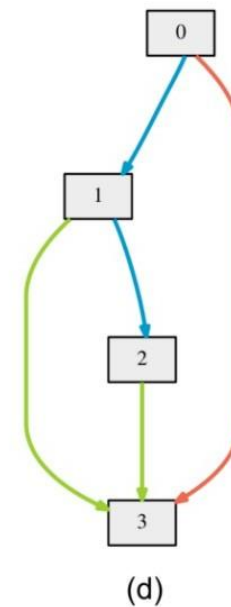
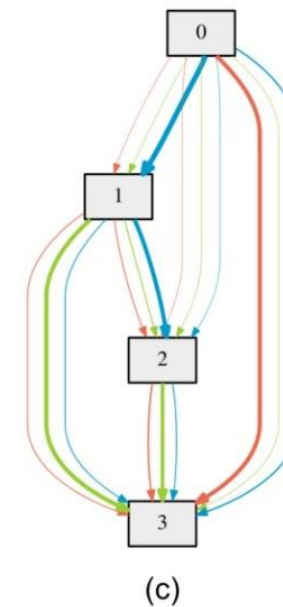
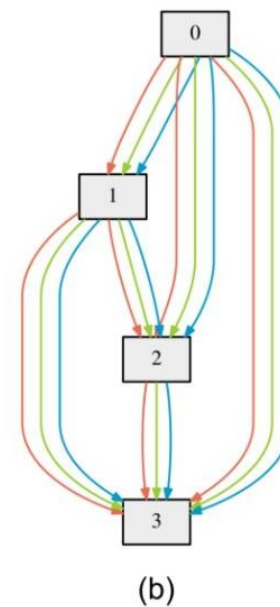
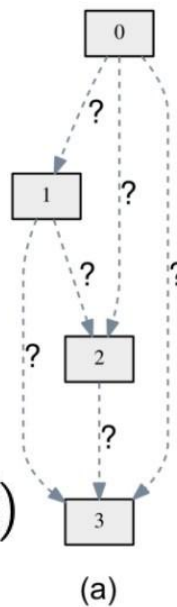
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3. Convert softmax to argmax



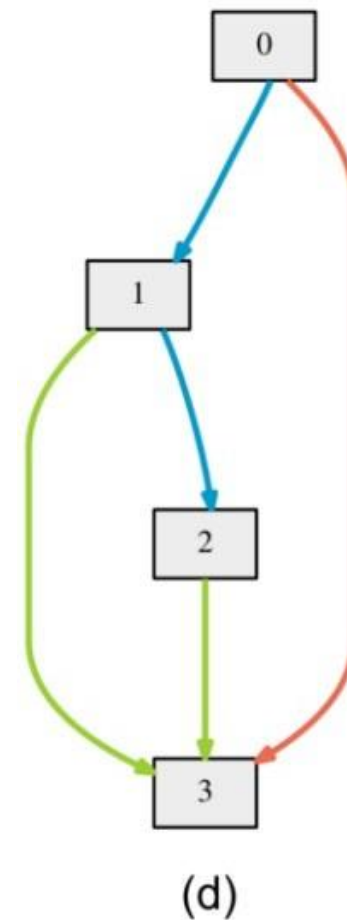
Faster but still non-trivial

Continuous Structure Discovery

In general, optimization over directed acyclic graphs (DAGs):

Graph $\mathbf{G}$, Data $\mathbf{X}$, Adjacency matrix $\mathbf{W}$:

$$\begin{array}{ccc} \min_W \ell(W; X) & \stackrel{?}{\iff} & \min_W \ell(W; X) \\ \text{s.t. } G(W) \in \text{DAG} & & \text{s.t. } h(W) = 0 \\ \text{(combinatorial 🤖)} & & \text{(smooth 😎)} \end{array}$$



Continuous Structure Discovery

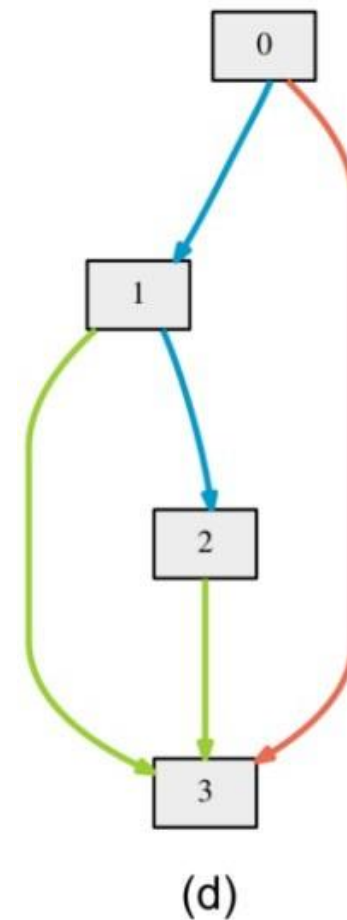
$$\begin{array}{ccc} \min_W \ell(W; X) & \stackrel{?}{\iff} & \min_W \ell(W; X) \\ \text{s.t. } G(W) \in \text{DAG} & & \text{s.t. } h(W) = 0 \end{array}$$

$$h(W) = \text{tr}(e^{W \circ W}) - d,$$

$$e^A = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \dots$$

- K -th power of adjacency matrix $\mathbf{W}$ counts the number of k -step paths from one node to another.
- If the diagonal of the matrix power is all zeros, there are no k -step cycles.
- Acyclic = check all $k = 1, 2, \dots$, size of graph.

Can now do continuous optimization to solve for W , but **nonconvex**



(d)

Continuous Structure Discovery

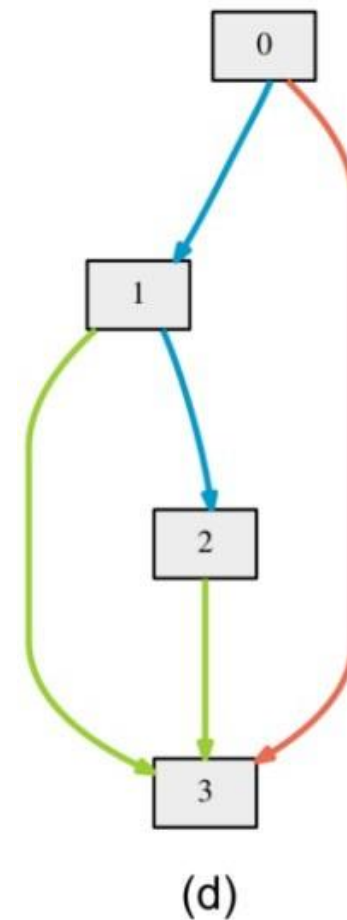
$$\begin{array}{ccc} \min_W \ell(W; X) & \stackrel{?}{\iff} & \min_W \ell(W; X) \\ \text{s.t. } G(W) \in \text{DAG} & & \text{s.t. } h(W) = 0 \end{array}$$

In our paper, we showed that such a function h exists,

$$h(W) = \text{tr}(e^{W \circ W}) - d,$$

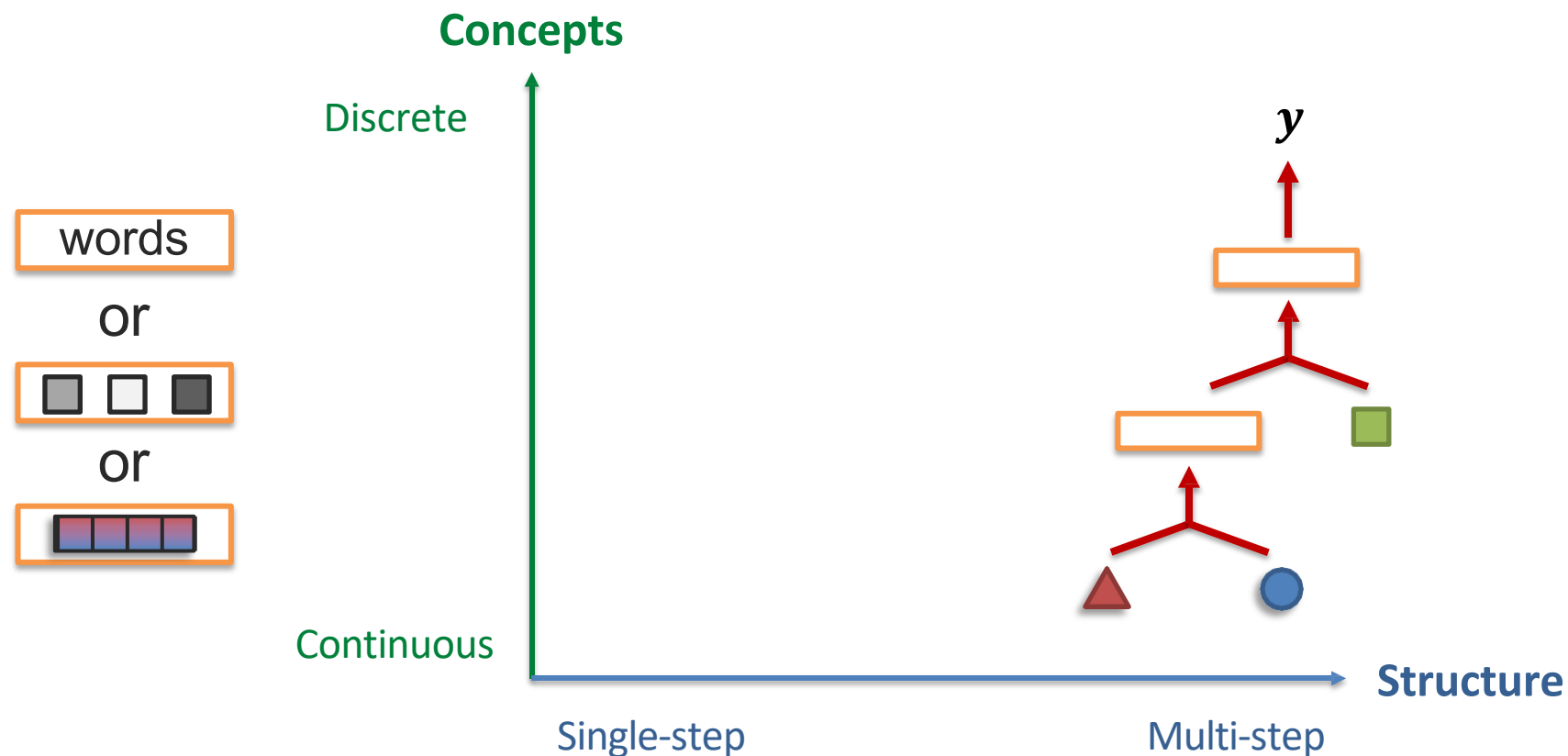
and that it has a simple gradient:

$$\nabla h(W) = (e^{W \circ W})^T \circ 2W.$$



Sub-Challenge 3b: Intermediate Concepts

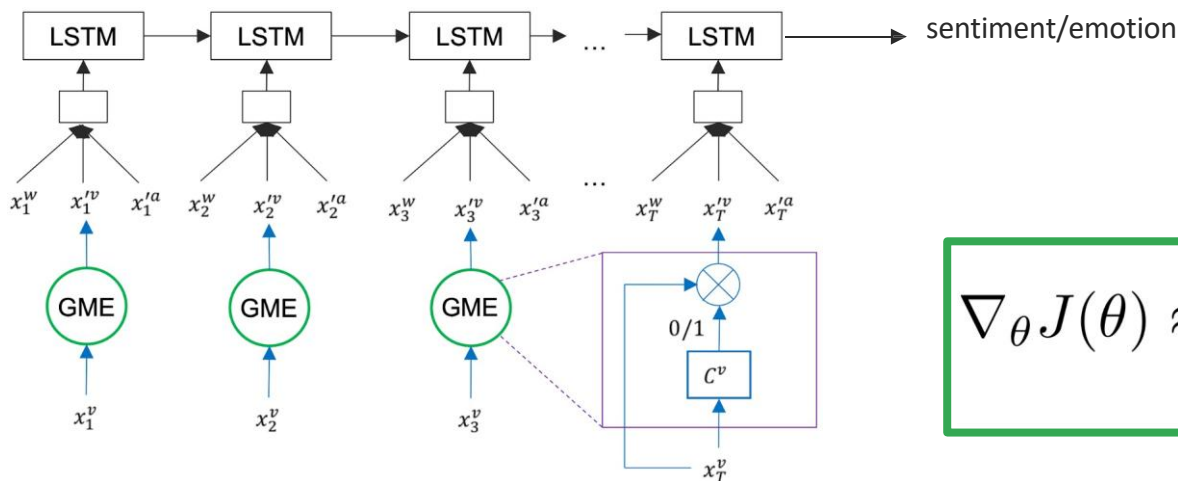
Definition: The parameterization of individual multimodal concepts in the reasoning process.



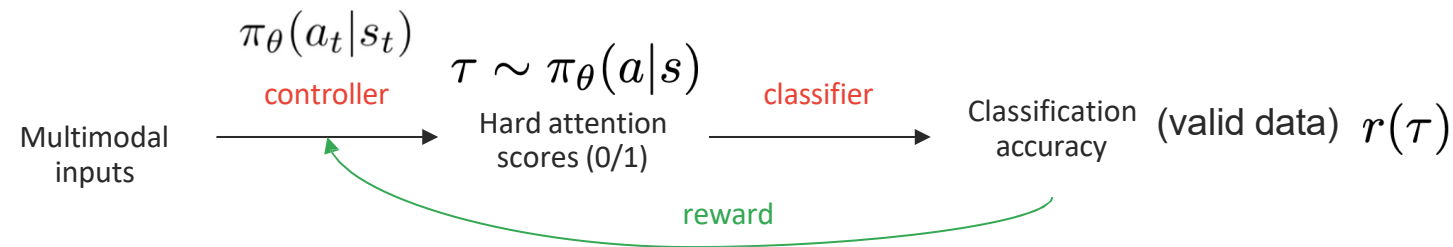
Discrete Concepts via Hard Attention

Hard attention 'gates' (0/1) rather than soft attention (softmax between 0-1)

- Can be seen as discrete layers in between differentiable neural net layers



$$\nabla_{\theta} J(\theta) \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

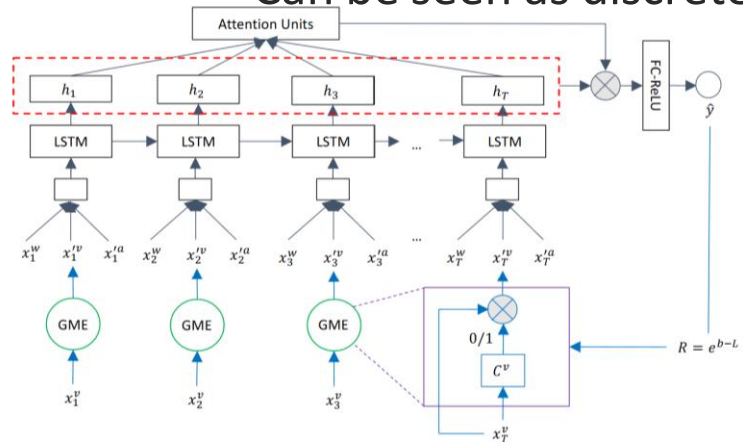


[Xu et al., Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. ICML 2015]

[Chen et al., Multimodal Sentiment Analysis with Word-level Fusion and Reinforcement Learning. ICMI 2017]

Discrete Concepts via Hard Attention

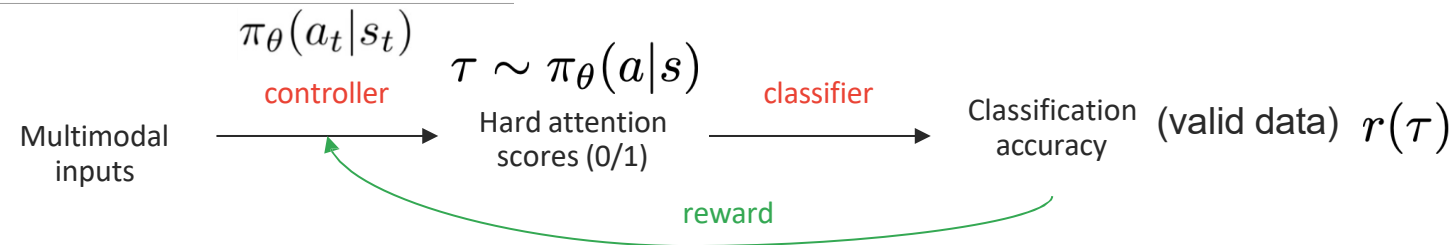
Hard attention 'gates' (0/1) rather than soft attention (softmax between 0-1)
 - Can be seen as discrete layers in between differentiable neural net layers



sentiment/emotion

$$\nabla_{\theta} J(\theta) \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

Figure 1: Architecture of the GME-LSTM(A) model for the visual modality. C^v is the controller for the visual modality that selectively allows visual inputs x_t^v to pass. FC-ReLU is a fully-connected layer with rectified linear unit (ReLU) as activation. After obtaining a sentiment prediction $\hat{y}$ and loss $\mathcal{L}$, we use $R = e^{b-\mathcal{L}}$ as the reward signal to train the visual input gate controller C^v .



[Xu et al., Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. ICML 2015]

[Chen et al., Multimodal Sentiment Analysis with Word-level Fusion and Reinforcement Learning. ICMI 2017]

Discrete Concepts via Hard Attention

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- Can be seen as discrete layers in between differentiable neural net layers

Sentiment analysis,
emotion recognition



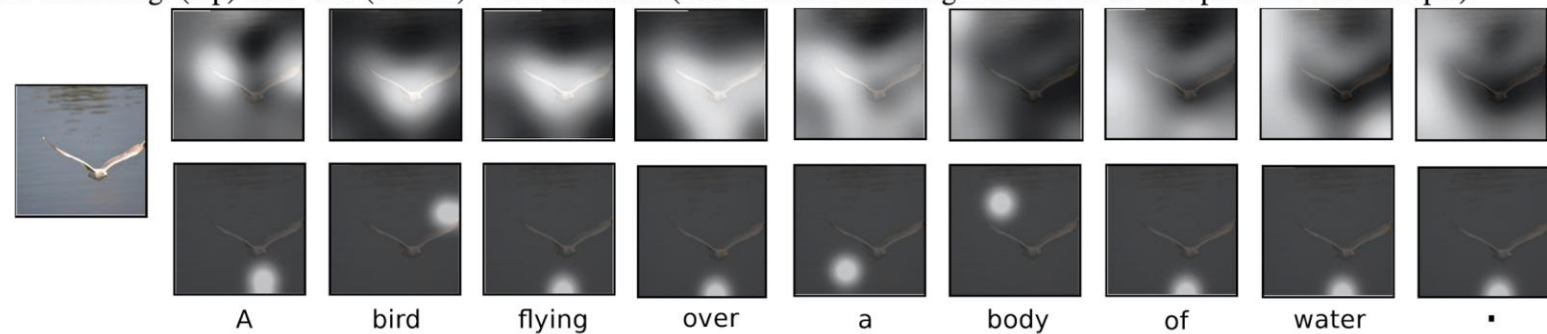
Reject

Pass

Reject

Image captioning

Figure 3. Visualization of the attention for each generated word. The rough visualizations obtained by upsampling the attention weights and smoothing. (top) “soft” and (bottom) “hard” attention (note that both models generated the same captions in this example).

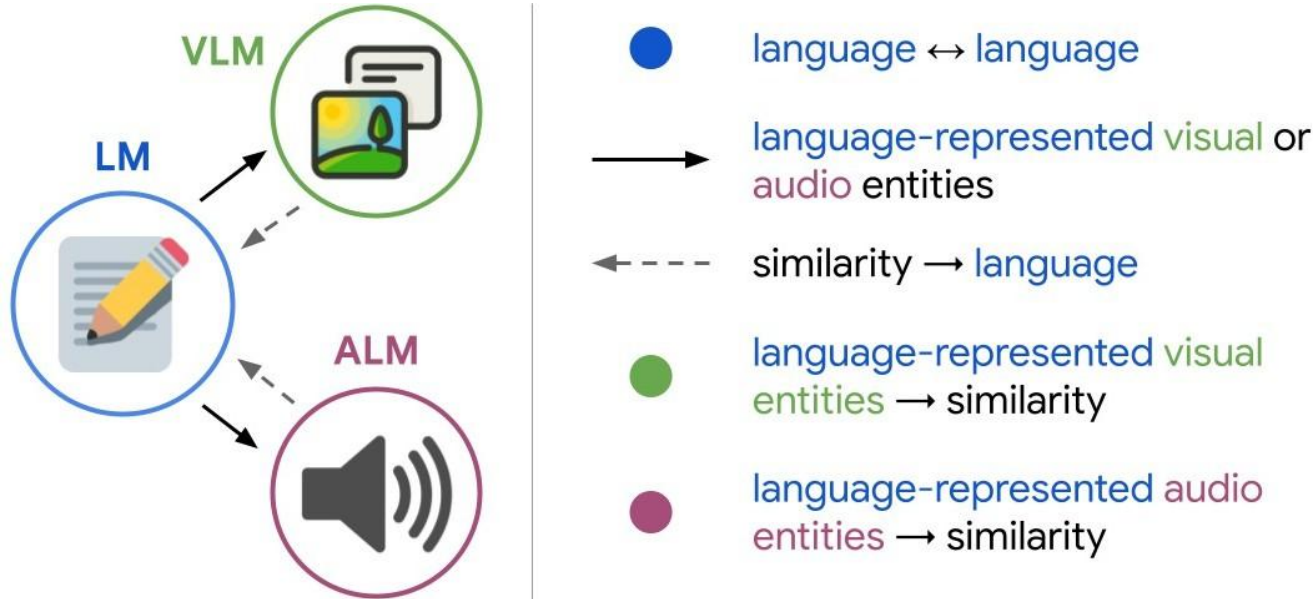


[Xu et al., Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. ICML 2015]

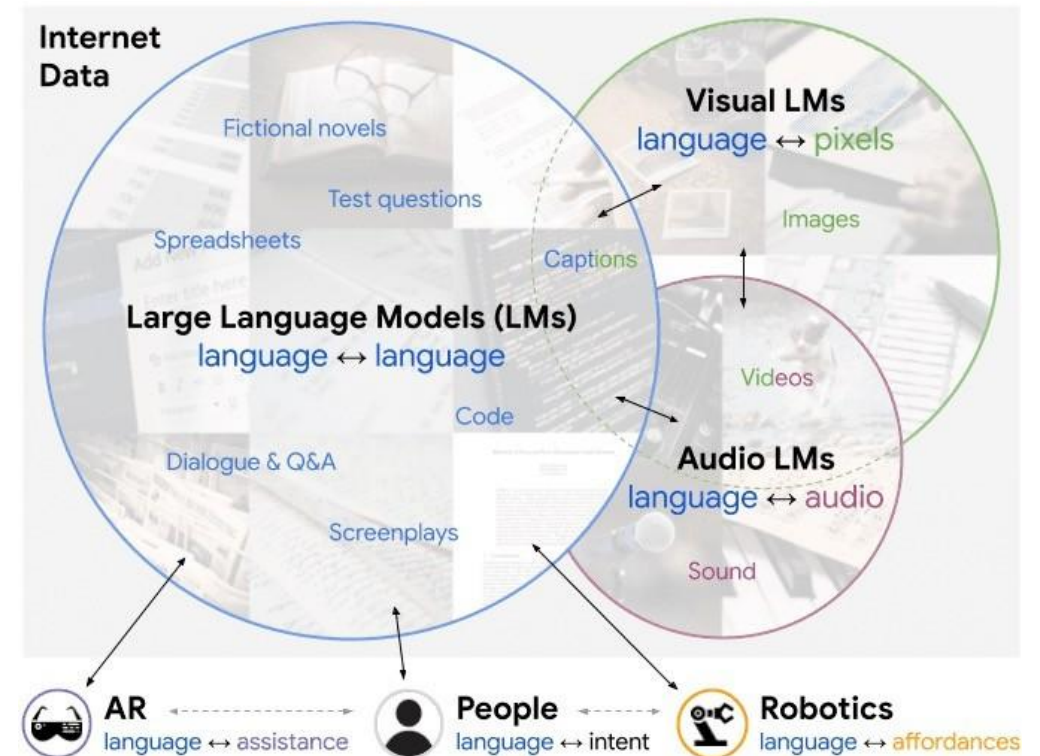
[Chen et al., Multimodal Sentiment Analysis with Word-level Fusion and Reinforcement Learning. ICMI 2017]

Discrete Concepts via Language

- Large language/video/audio models interacting with each other
- Each language model has its own distinct *domain knowledge*
- Interaction is scripted and zero-shot



Guided multimodal discussion



Combining domain knowledge

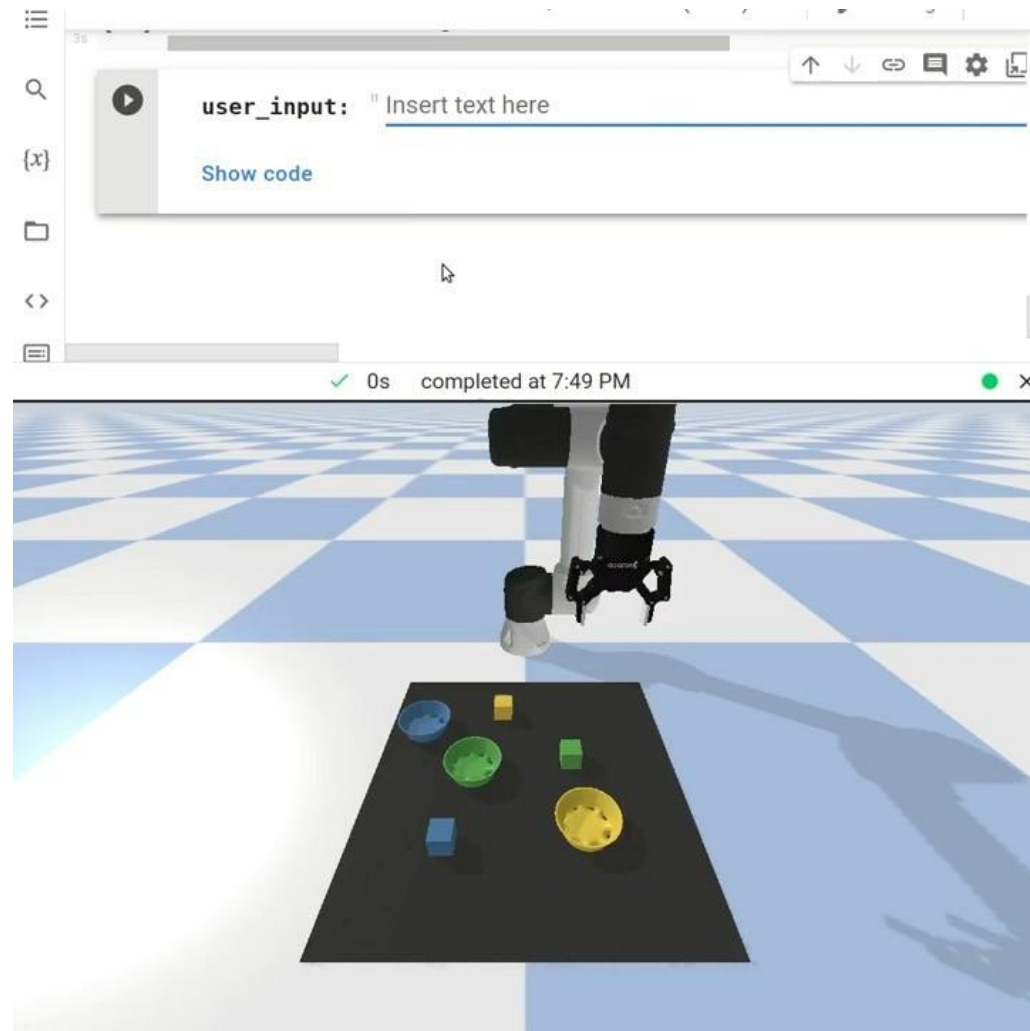
Discrete Concepts via Language

Image captioning

Zero-Shot
Socratic
Internet
Image
Captioning

Discrete Concepts via Language

Robot perception and planning



[Zeng et al., Socratic Models: Composing Zero-Shot Multimodal Reasoning with Language. arXiv 2022]

Discrete Concepts via Language

Video reasoning



11:09 AM: Places: **living room**. Objects: **remote control**, **television**, **netflix**. Commonsense activities: **watching netflix**. Most likely: **watching netflix**. I was **watching netflix**.

Language-based World-state History

8:31 AM: Places: clean room. Objects: shorts, jeans, shirt. Commonsense activities: getting dressed. Most likely: getting dressed. I was getting dressed.

10:17 AM: Places: kitchen. Objects: coffeemaker, waffle iron, kettle. Commonsense activities: making coffee, making waffles. Most likely: making coffee. Summary: I was making coffee.



Contextual Reasoning Q&A

Q: Why did I go to the front porch today?

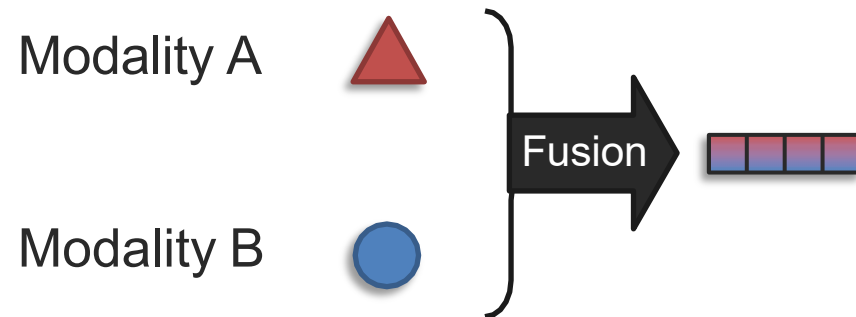
A: I went to the front porch today to receive a package.

Explanation: I saw on the porch a package and knew that I was expecting it.

Sub-Challenge 3c: Inference Paradigm

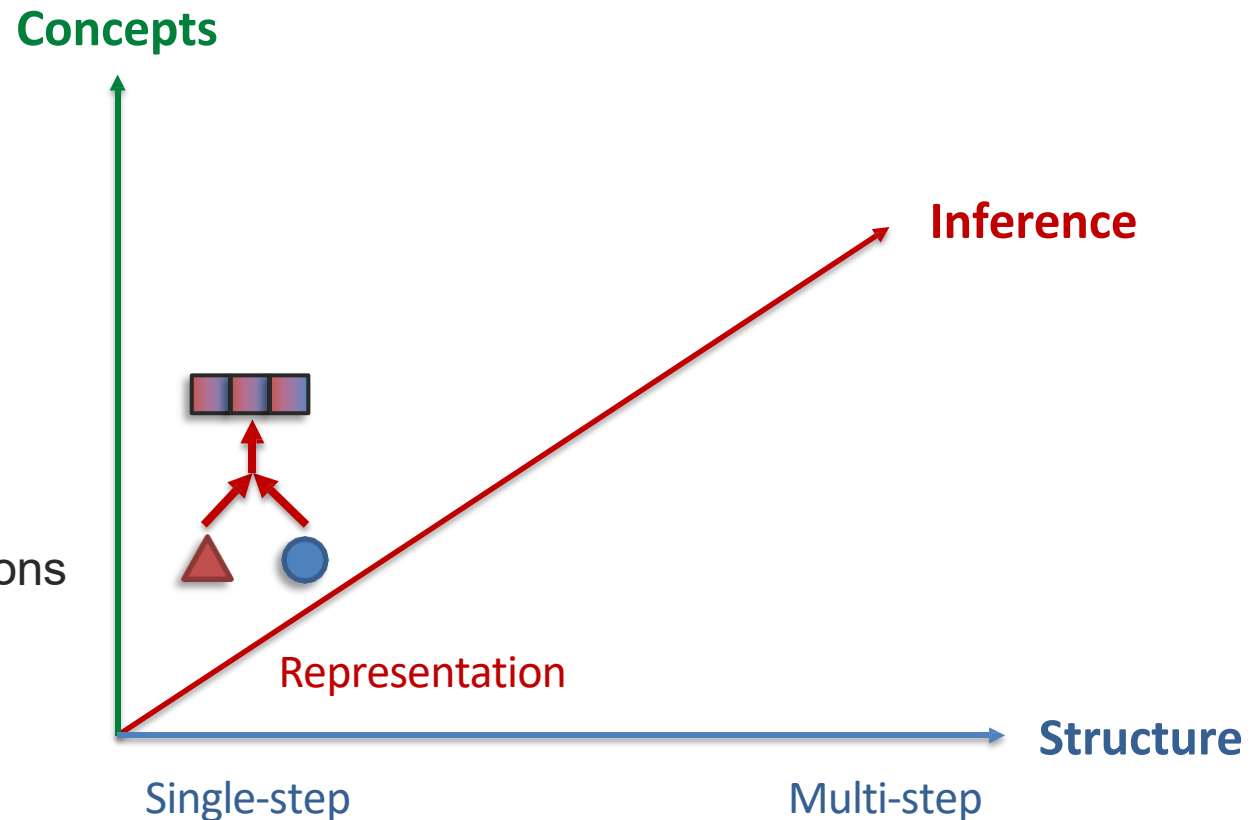
Definition: How increasingly abstract concepts are inferred from individual multimodal evidences.

Recall representation fusion:



Potential issues:

- Models may capture spurious correlations
- Not robust to targeted manipulations
- Lack of interpretability/control

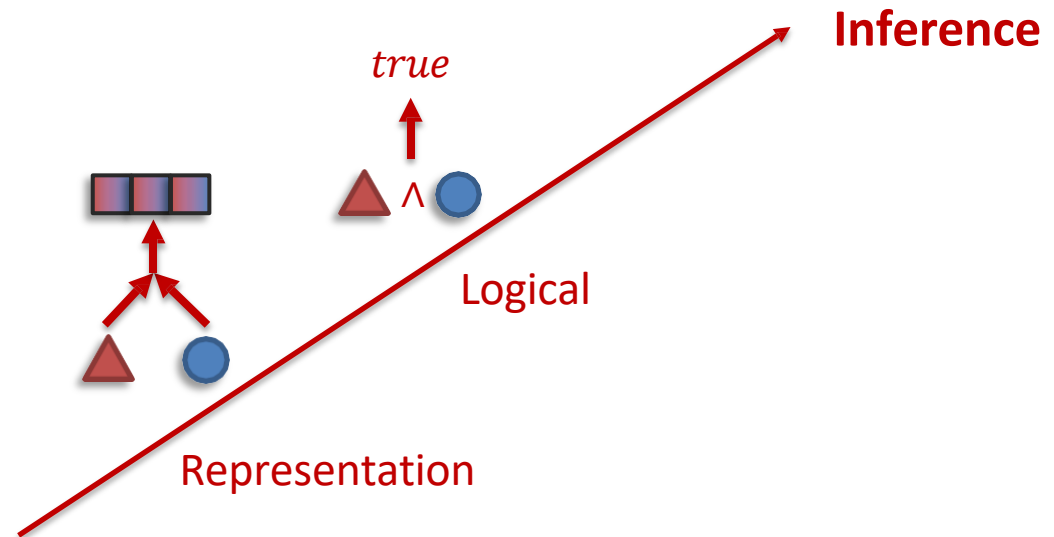


Sub-Challenge 3c: Inference Paradigm

Definition: How increasingly abstract concepts are inferred from individual multimodal evidences.

Towards explicit inference paradigms:

1. Logical inference: given premises inferred from multimodal evidence, how can one derive **logical** conclusions?



Logical Inference

Recall error
analysis!

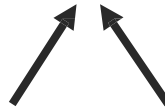
Inference through logical operators in question



*Is there beer AND is there a
WINE GLASS?*



*Is the man NOT wearing
shoes AND is there beer?*



Is there beer?

Is the man wearing shoes?

Adversarial
contrasts/foils



Logical connectives



Basic premises



Existing models struggle to capture logical connectives.
How can we make them more logical?

Logical Inference

Inference through logical operators in question

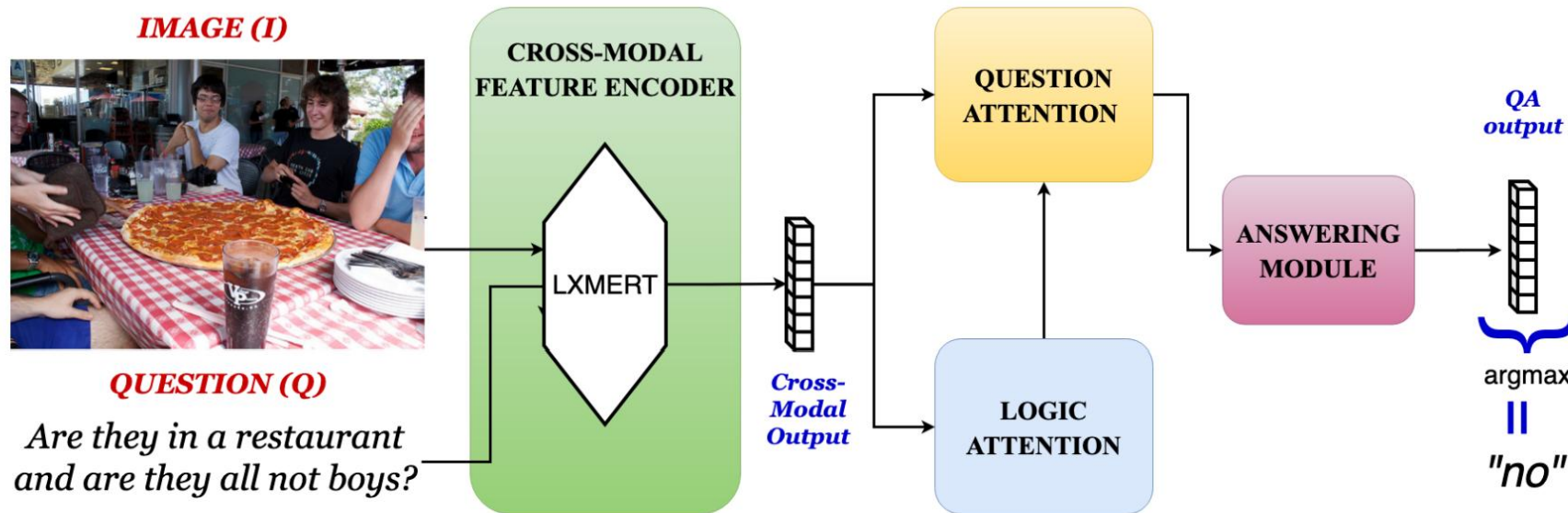


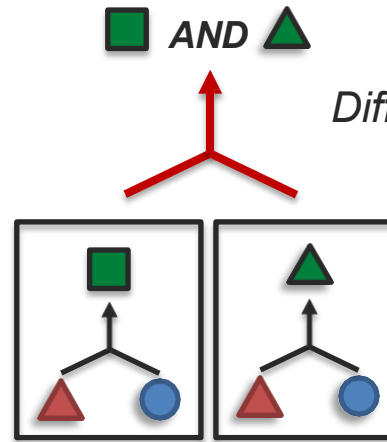
Fig. 3: LOL model architecture showing a cross-modal feature encoder followed by our Question-Attention (q_{ATT}) and Logic Attention (ℓ_{ATT}) modules. The concatenated output of is used by the Answering Module to predict the answer.

Logical Inference

Inference through logical operators in question



Are they in a restaurant **AND** are they all boys?



Differentiable **AND** composition operator!

Also applies to other logic connectives:
AND, OR, NOT



Are they in a restaurant?



Are they all boys?

Soft Logical Operators

Inference through logical operators in question

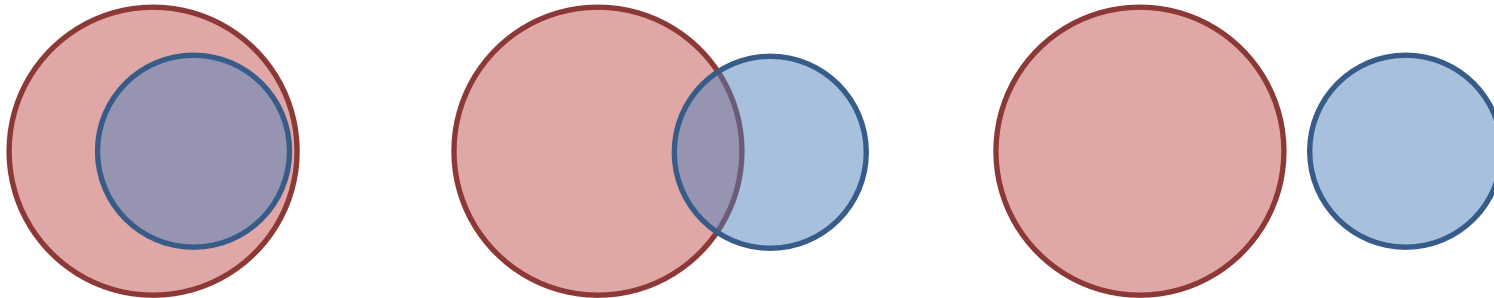
Fréchet inequalities to make logical functions differentiable:

- Probability of an **intersection** of events

$$\max(0, \mathbb{P}(A) + \mathbb{P}(B) - 1) \leq \mathbb{P}(A \cap B) \leq \min(\mathbb{P}(A), \mathbb{P}(B)),$$

- Probability of a **union** of events

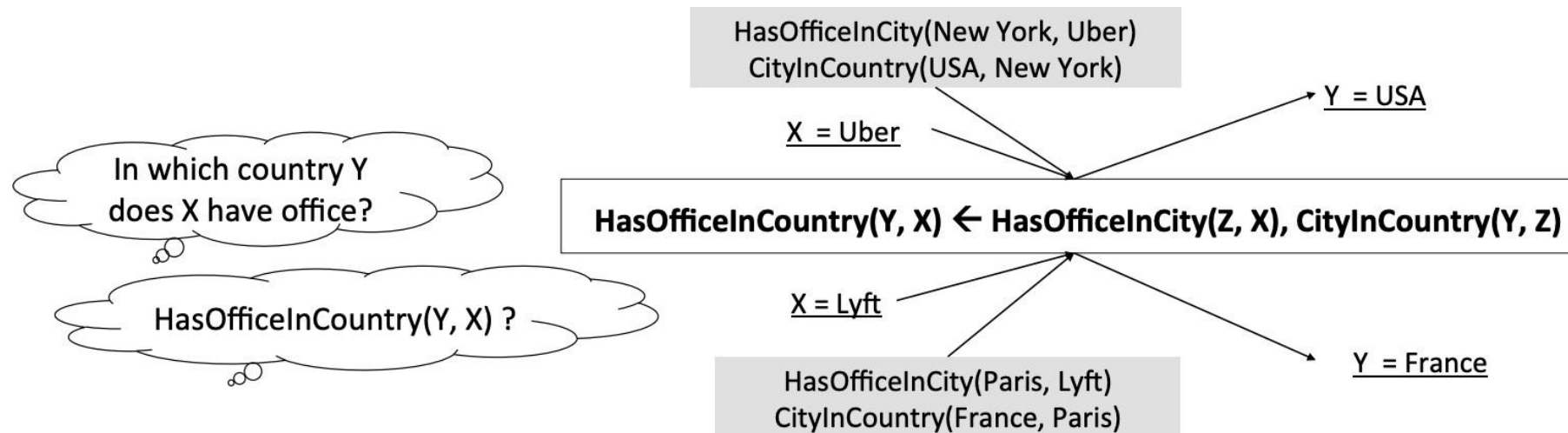
$$\max(\mathbb{P}(A), \mathbb{P}(B)) \leq \mathbb{P}(A \cup B) \leq \min(1, \mathbb{P}(A) + \mathbb{P}(B)).$$



Logical Inference Challenges

Open
challenges

Many open directions



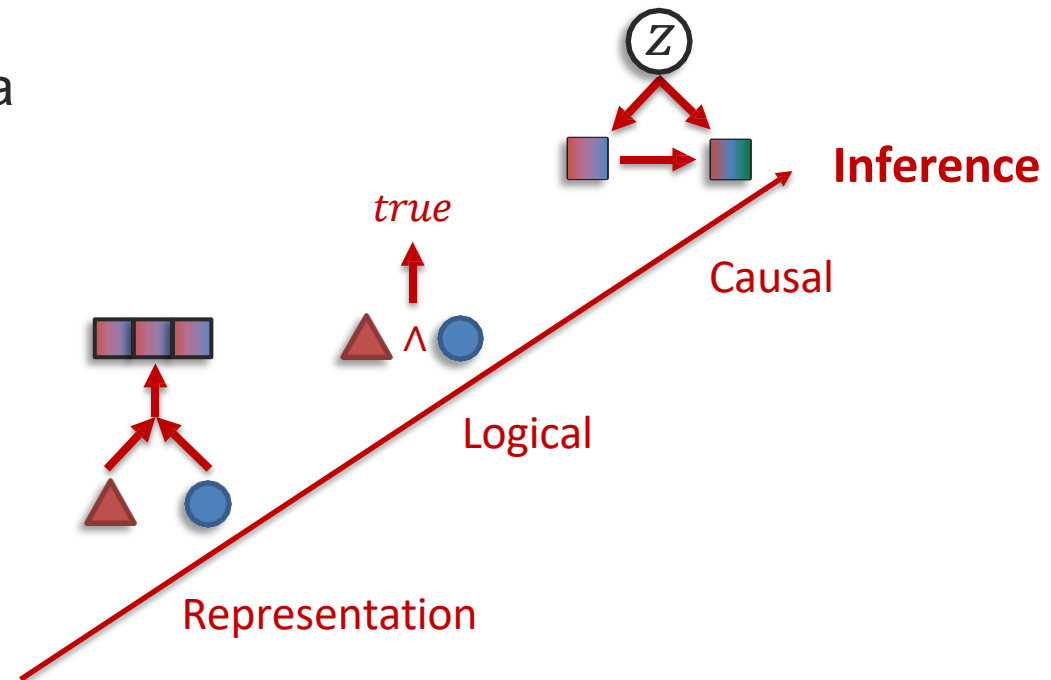
Differentiable knowledge base reasoning

Sub-Challenge 3c: Inference Paradigm

Definition: How increasingly abstract concepts are inferred from individual multimodal evidences.

Towards explicit inference paradigms:

1. Logical inference
2. Causal inference: how can one determine the actual **causal** effect of a variable in a larger system?



Causal Inference

Intervention

Causal inference is reliant on the idea of interventions —what outcome might have occurred if X happened (an intervention), possibly contrary to observed data.

vs association describes how things are. Causation describes how things would have been under different circumstances.

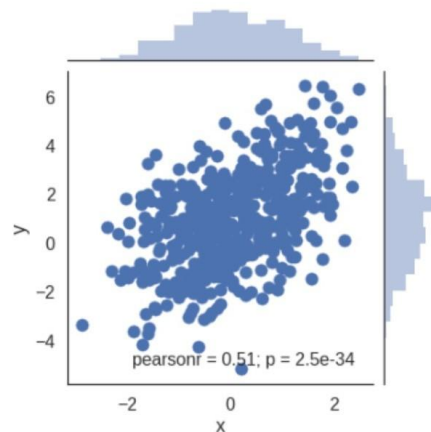
(side note: correlation is a specific type of linear association)

Causal Inference

Intervention

Causal inference is reliant on the idea of interventions —what outcome might have occurred if X happened (an intervention), possibly contrary to observed data.

```
x = randn()  
y = x + 1 + sqrt(3)*randn()
```

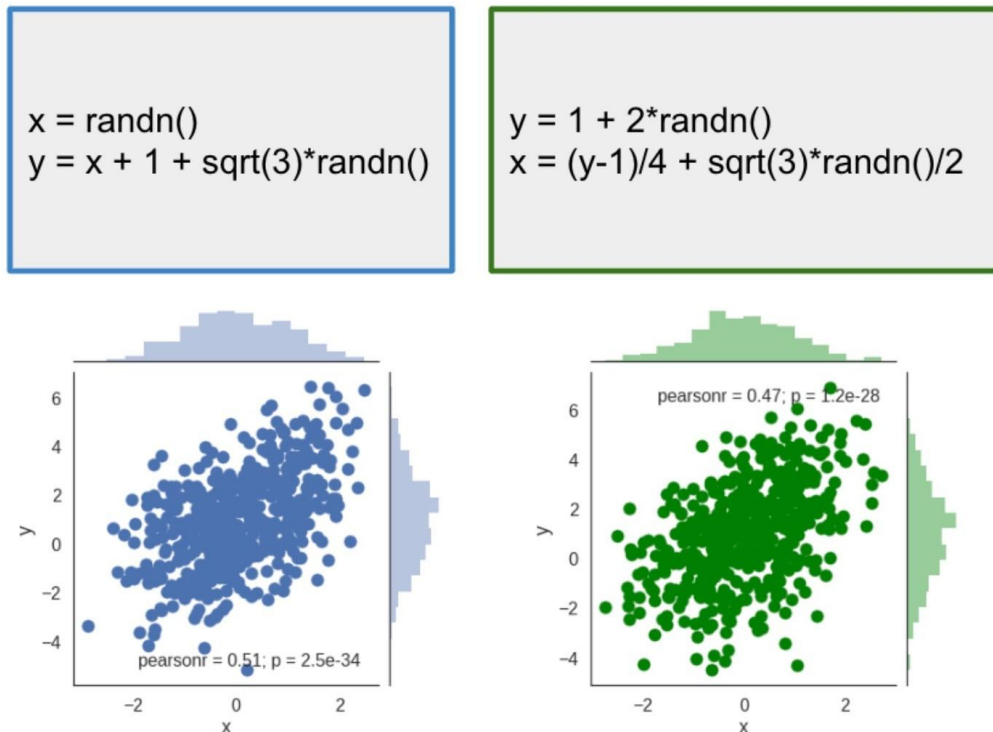


[Example from Ferenc Huszár: <https://www.inference.vc/causal-inference-2-illustrating-interventions-in-a-toy-example/>]

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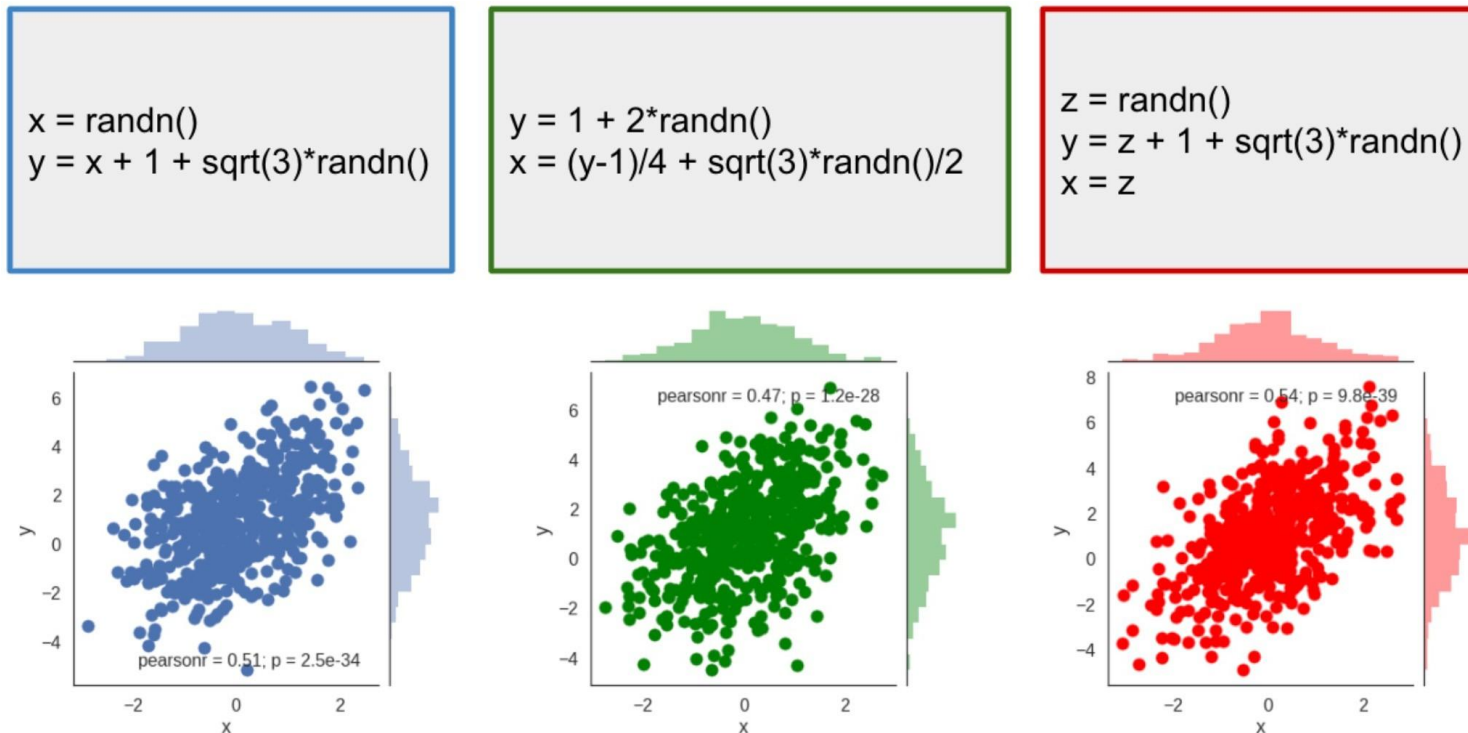


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Causal Inference

Intervention

Let's say I really want to set the value of x to 3.

```
x = randn()  
x = 3  
y = x + 1 + sqrt(3)*randn()  
x = 3
```

```
y = 1 + 2*randn()  
x = 3  
x = (y-1)/4 + sqrt(3)*randn()/2  
x = 3
```

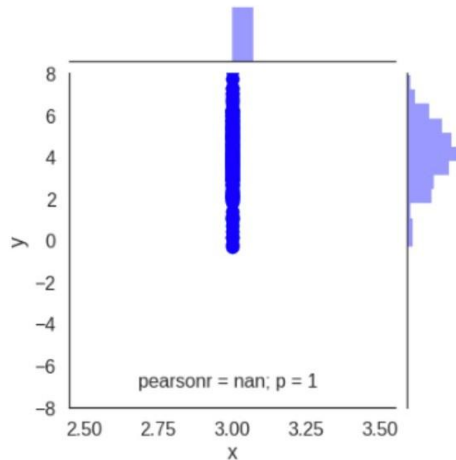
```
z = randn()  
x = 3  
x = z  
x = 3  
y = z + 1 + sqrt(3)*randn()  
x = 3
```

Causal Inference

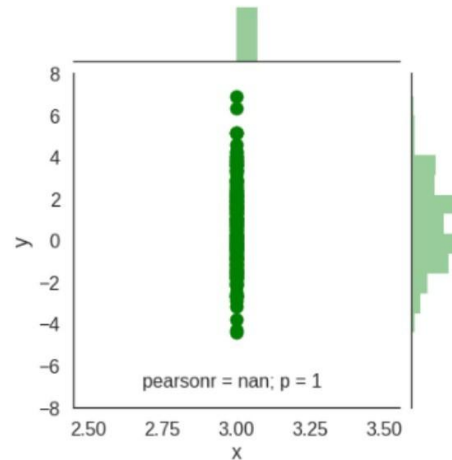
Intervention

Let's say I really want to set the value of x to 3. What happens to y ?

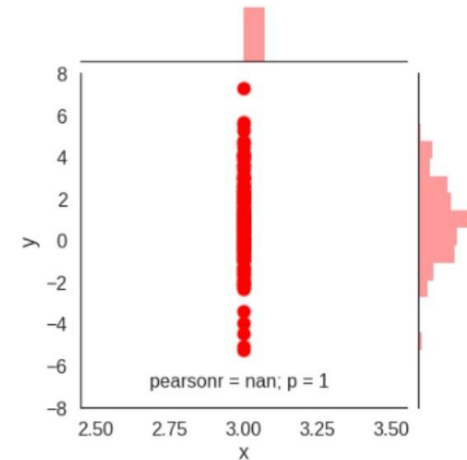
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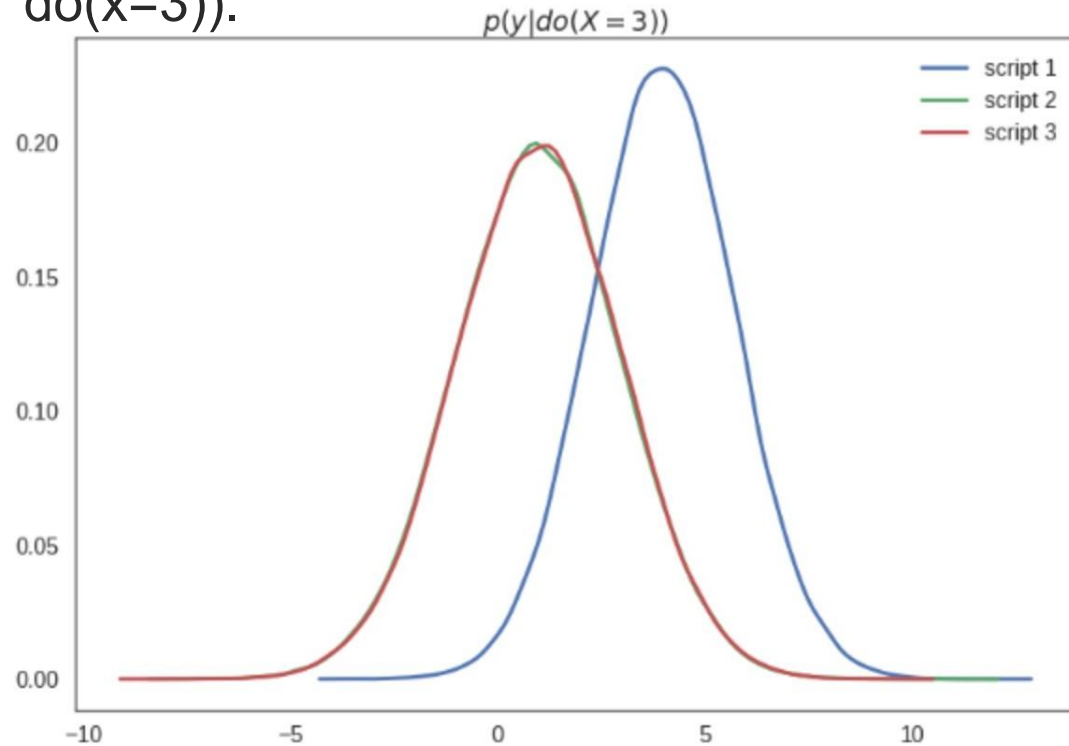
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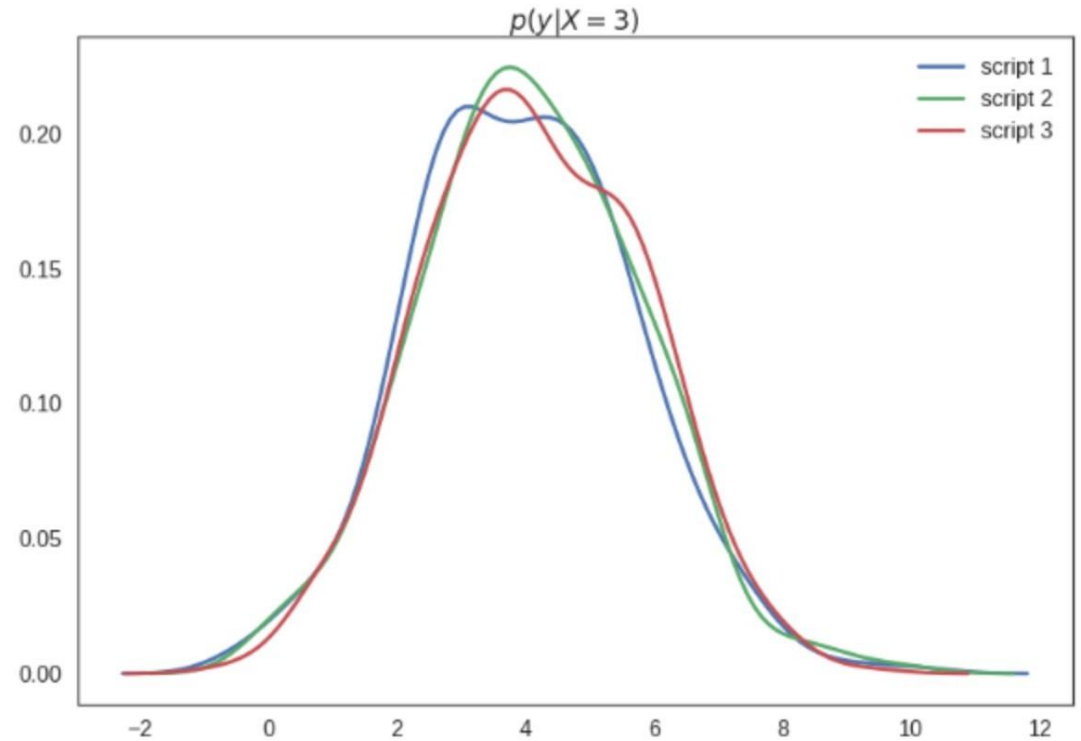
Causal Inference

Intervention

Interventional distribution of y : $p(y \mid \text{do}(x=3))$.



Conditional distribution of y : $p(y \mid x=3)$.



The joint distribution of data alone is insufficient to predict behavior under interventions.

[Example from Ferenc Huszár: <https://www.inference.vc/causal-inference-2-illustrating-interventions-in-a-toy-example/>]

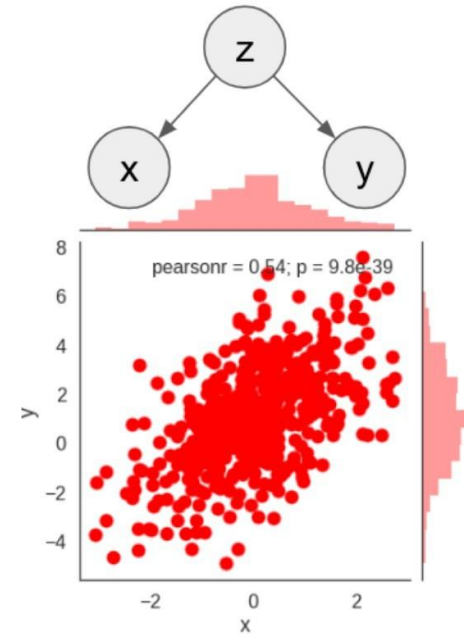
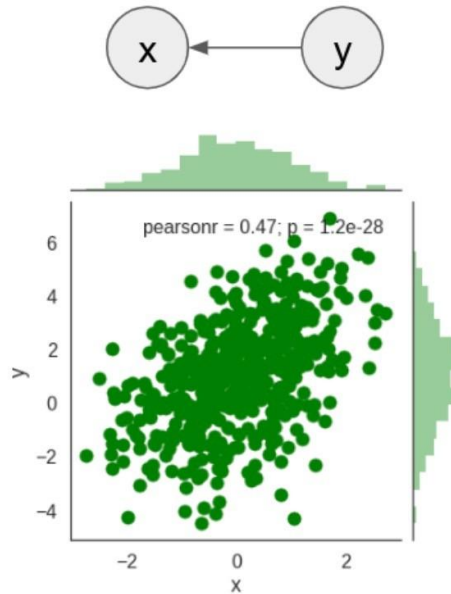
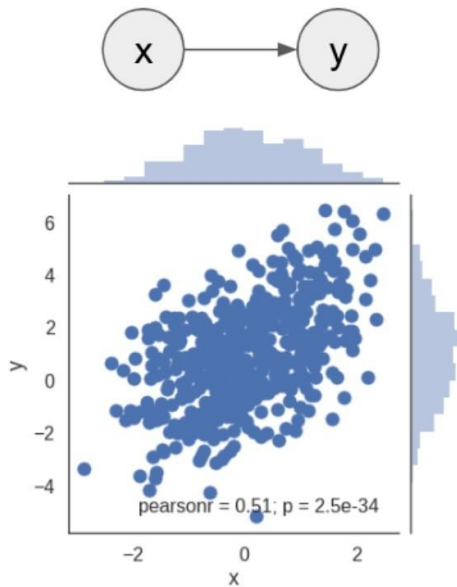
Causal Inference

Causal diagrams: arrow pointing from cause to effect.

```
x = randn()
y = x + 1 + sqrt(3)*randn()
```

```
y = 1 + 2*randn()
x = (y-1)/4 + sqrt(3)*randn()/2
```

```
z = randn()
y = z + 1 + sqrt(3)*randn()
x = z
```



Causal Inference

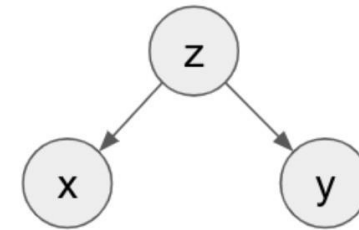
Intervention mutilates the graph by removing all edges that point into the variable on which intervention is applied (in this case x).



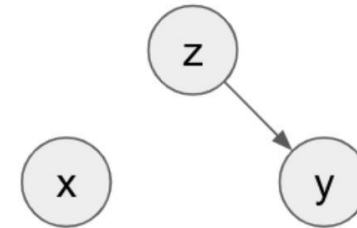
$$P(y|do(X)) = p(y|x)$$



$$P(y|do(X)) = p(y)$$



$$P(y|do(X)) = p(y)$$



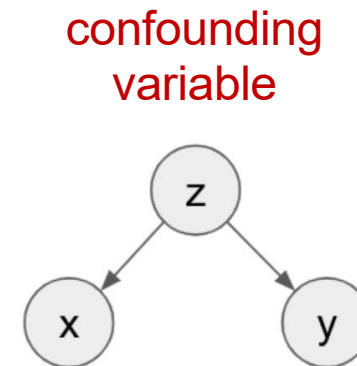
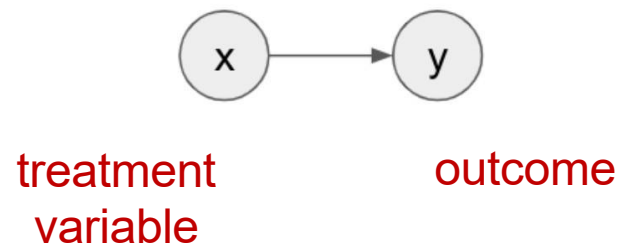
Causal Inference

Intervention in real-life is typically very hard!

E.g., does treatment x treat disease y ?

Can I estimate the intervention $p(y|\text{do}(X=x))$?

Requires answering: all else being equal, what would be the patient's outcome if they had not taken the treatment?



Lots of work, see Judea Pearl, The Book of Why

Causal Inference

Causal VQA: does my multimodal model capture causation or correlation?

Covariant VQA

Target object in question

Q: How many zebras are there in the picture?

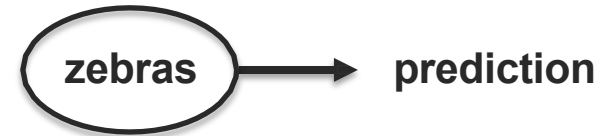
A: 2



Baselines:

2

i.e., treatment
variable



BUT: correlation or causation?

Causal Inference

Recall error analysis!

Causal VQA: does my multimodal model capture causation or correlation?

Covariant VQA

Target object in question

Q: How many zebras are there in the picture?

A: 2

zebra removed A: 1

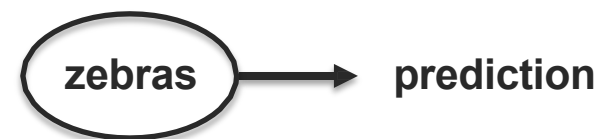


Baselines:

2

2

i.e., treatment variable



Interventional conditional: $p(y|do(zebras = 1))$

Existing models struggle to adapt to targeted causal interventions.
How can we make them more robust to spurious correlations?

Causal Inference

Causal VQA: does my multimodal model capture causation or correlation?

Invariant VQA

Target irrelevant object

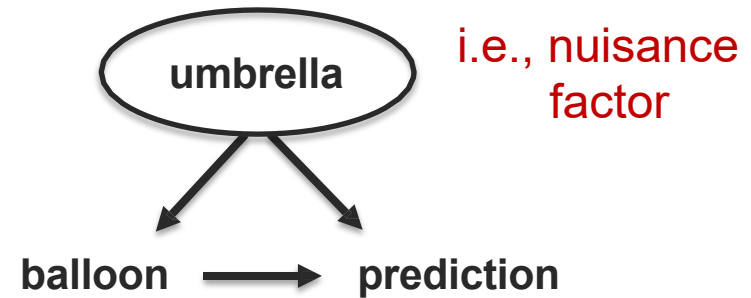
Q: What color is the balloon?

A: red



Baselines:

pink



Is my model picking up irrelevant objects?

Causal Inference

Recall error analysis!

Causal VQA: does my multimodal model capture causation or correlation?

Invariant VQA

Target irrelevant object

Q: What color is the balloon?

A: red

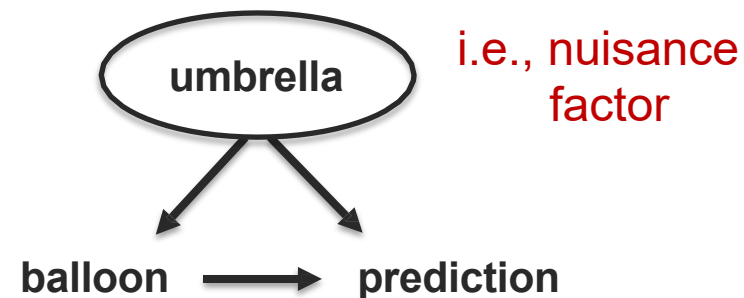
umbrellas removed; A: red



Baselines:

pink

red

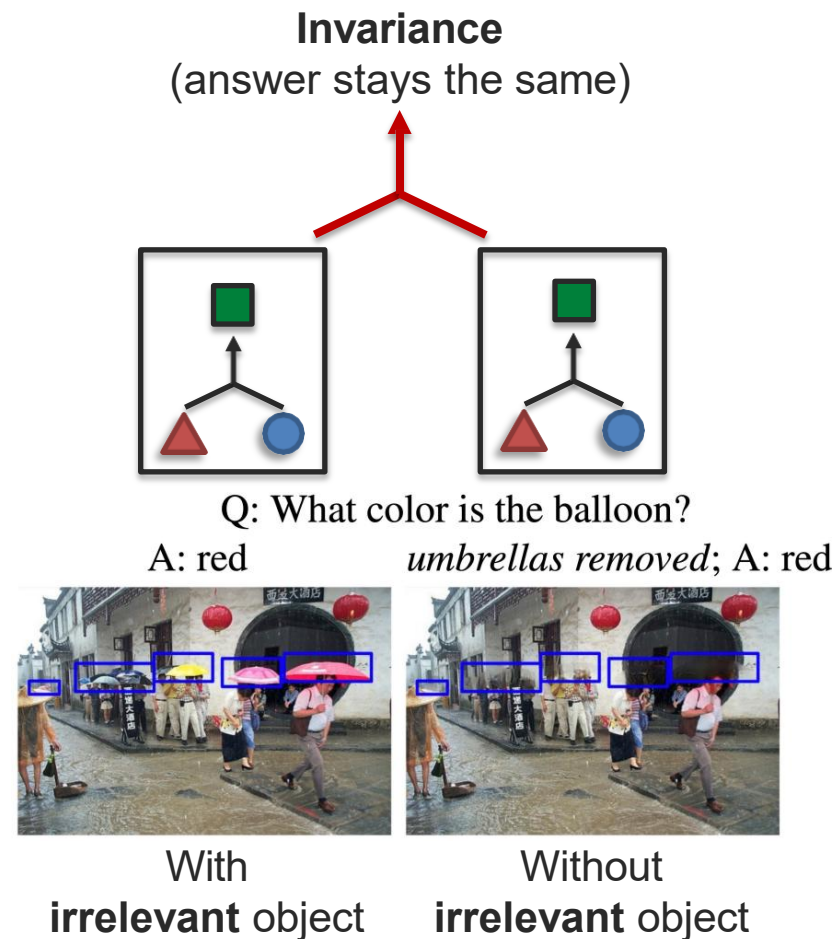
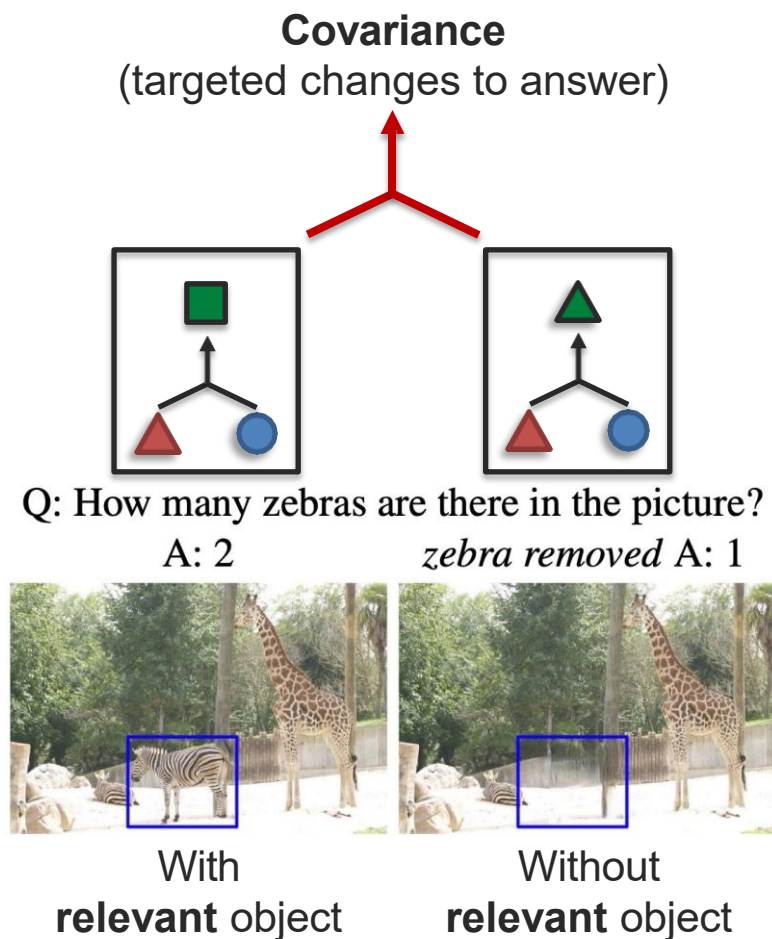


Interventional conditional: $p(y|do(no\ umbrella))$

Existing models struggle to adapt to targeted causal interventions.
How can we make them more robust to spurious correlations?

Causal Inference

Causal inference via data augmentation



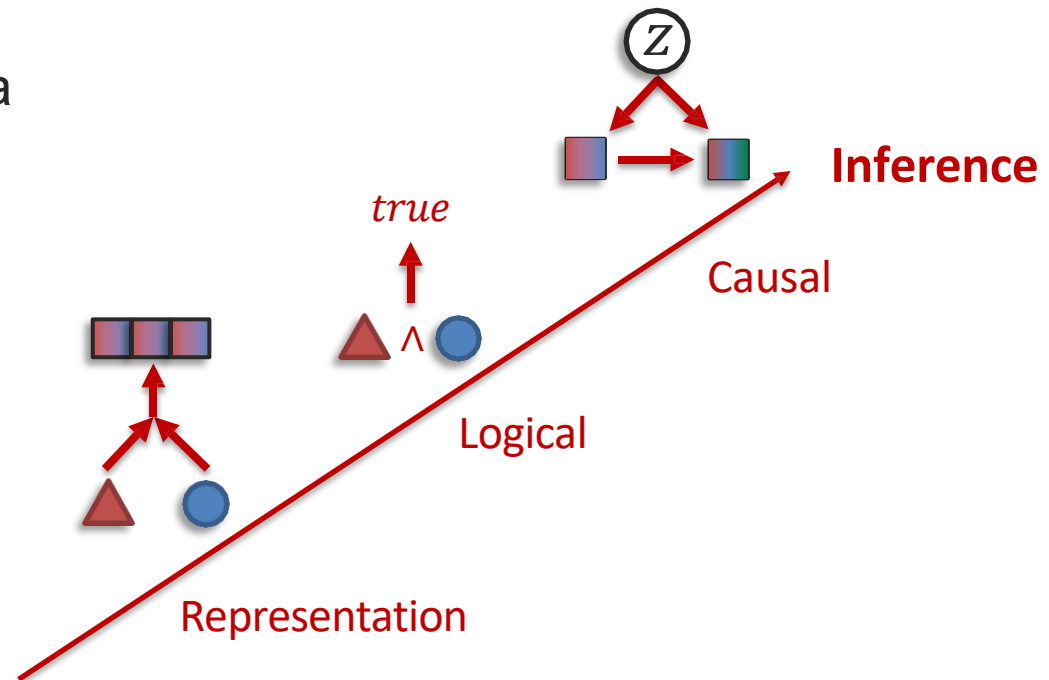
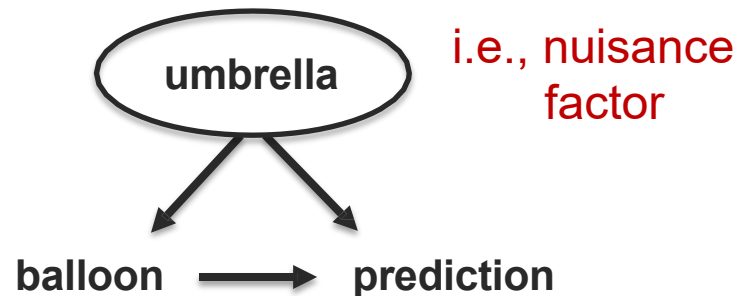
Sub-Challenge 3c: Inference Paradigm

Definition: How increasingly abstract concepts are inferred from individual multimodal evidences.

Towards explicit inference paradigms:

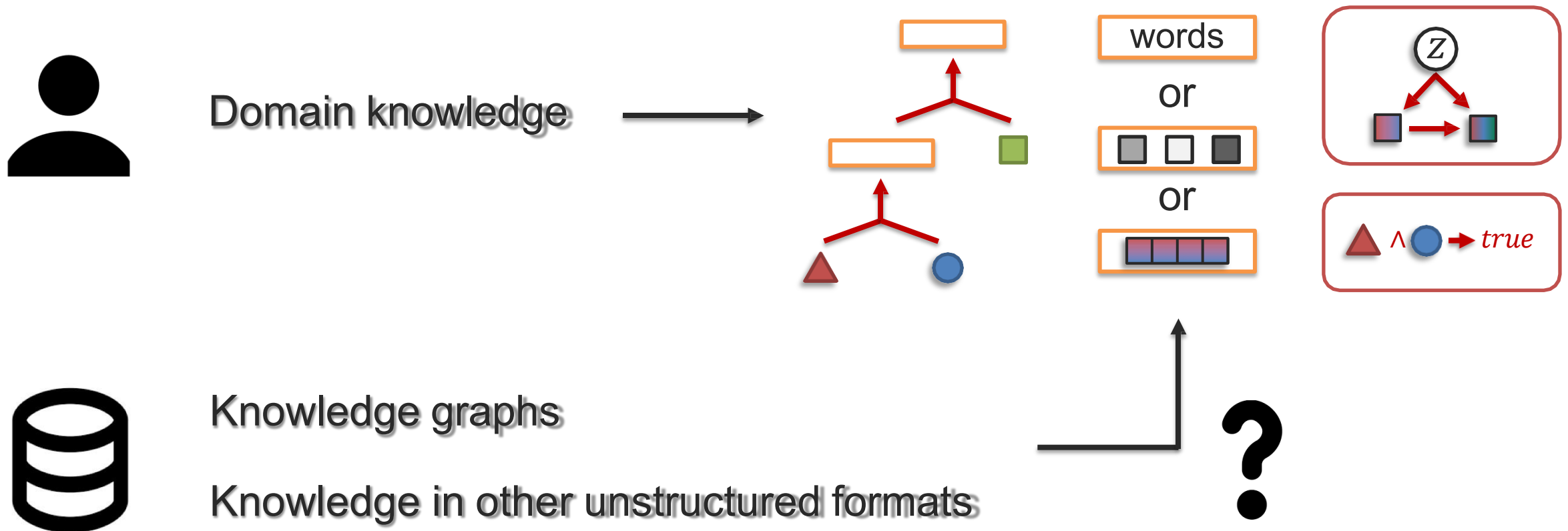
1. Logical inference
2. Causal inference: how can one determine the actual **causal** effect of a variable in a larger system?

Nice, but you don't get these for free!



Sub-Challenge 3d: Knowledge

Definition: The derivation of knowledge in the study of inference, structure, and reasoning.



External Knowledge: Multimodal Knowledge Graphs

Knowledge can also be gained from external sources



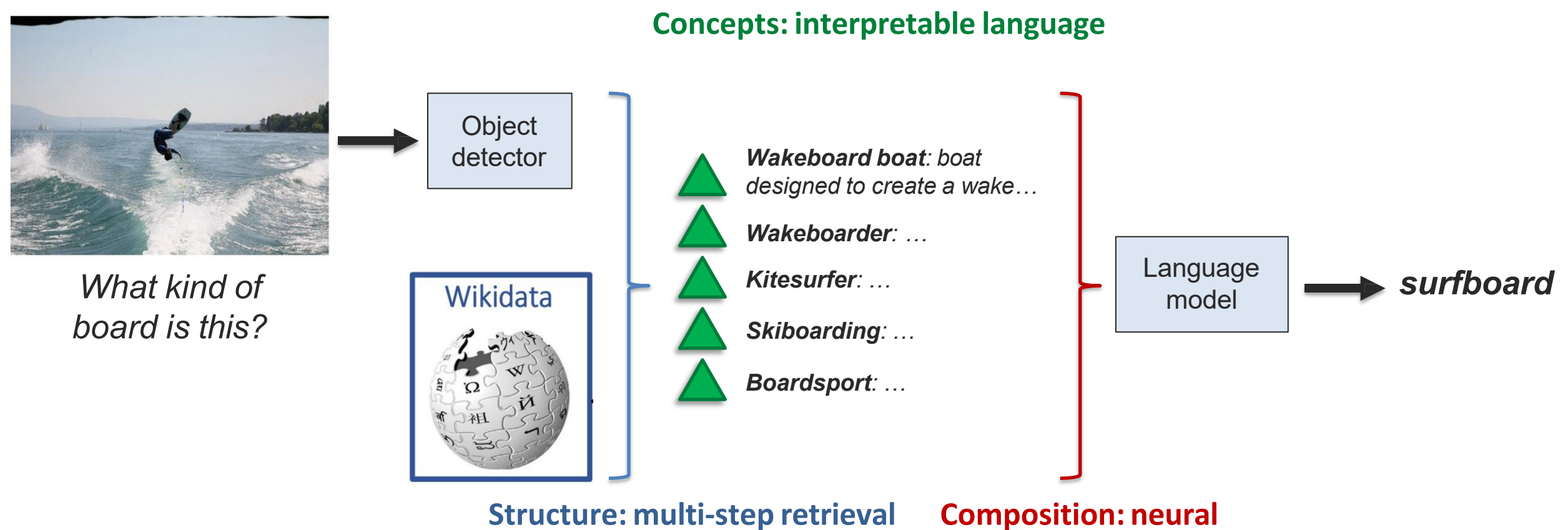
*What kind of
board is this?*

*Requires knowledge of water
sports, sports equipment, etc.*

Existing models struggle when external knowledge is needed.
How can we leverage external knowledge?

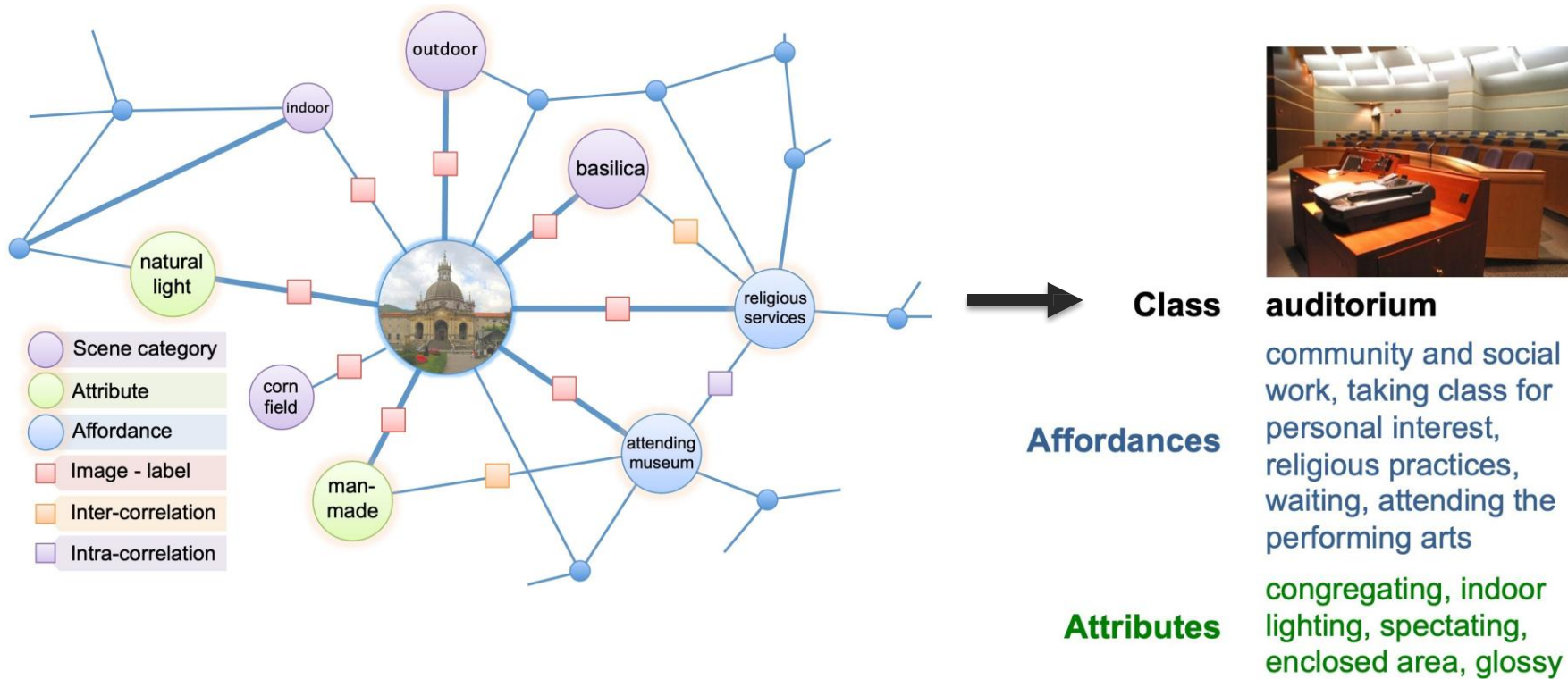
External Knowledge: Multimodal Knowledge Graphs

Knowledge can also be gained from external sources



External Knowledge: Multimodal Knowledge Graphs

Knowledge can also be gained from external sources



Concepts: interpretable
Structure: multi-step inference
Composition: graph-based

External Knowledge Challenges

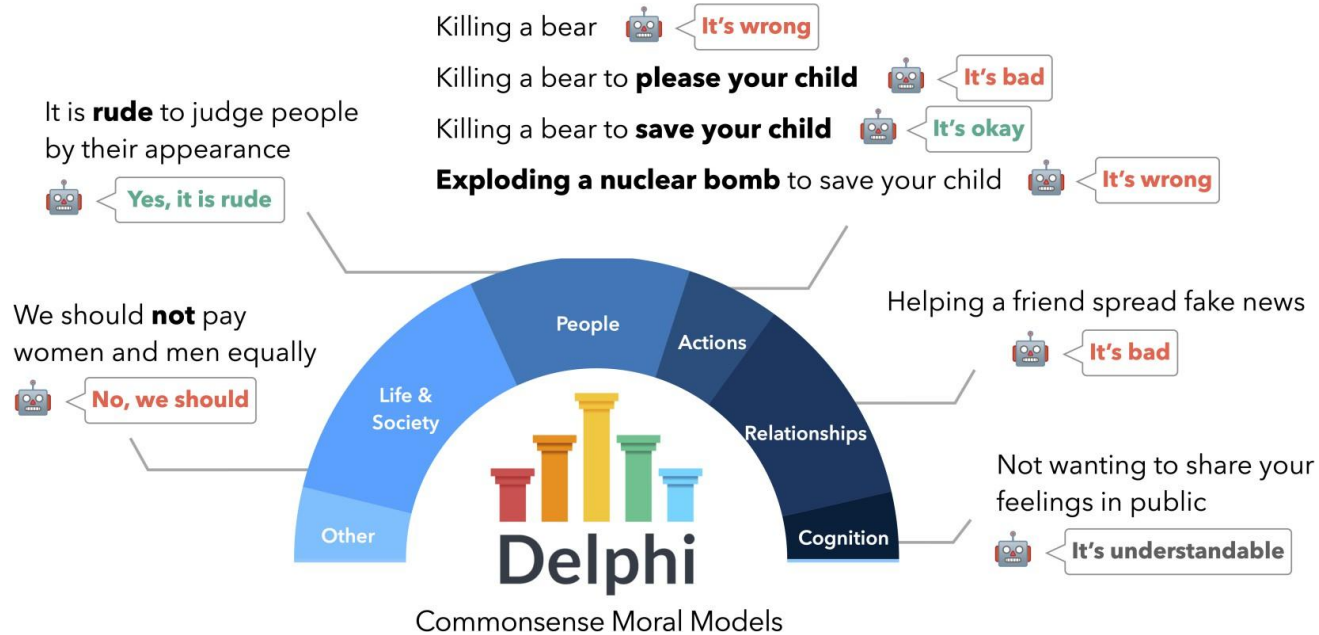
Open
challenges



Atomic: If-then commonsense

External Knowledge Challenges

Open challenges



Delphi: Moral commonsense



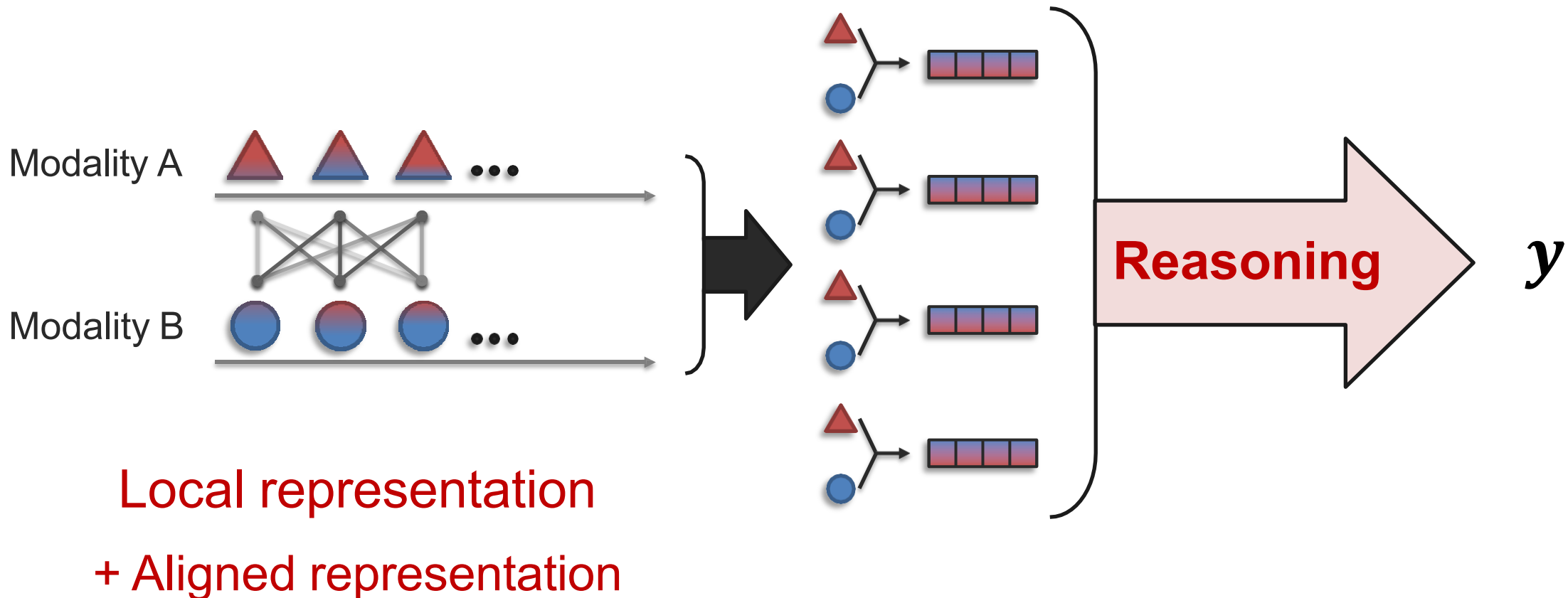
Social Chemistry: Social commonsense

[Jiang et al., Can Machines Learn Morality? The Delphi Experiment. arXiv 2021]

[Forbes et al., Social Chemistry 101: Learning to Reason about Social and Moral Norms. EMNLP 2020]

Summary: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.



The Challenge of Compositionality

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.



(a) some plants
surrounding a
lightbulb



(b) a lightbulb surrounding some plants

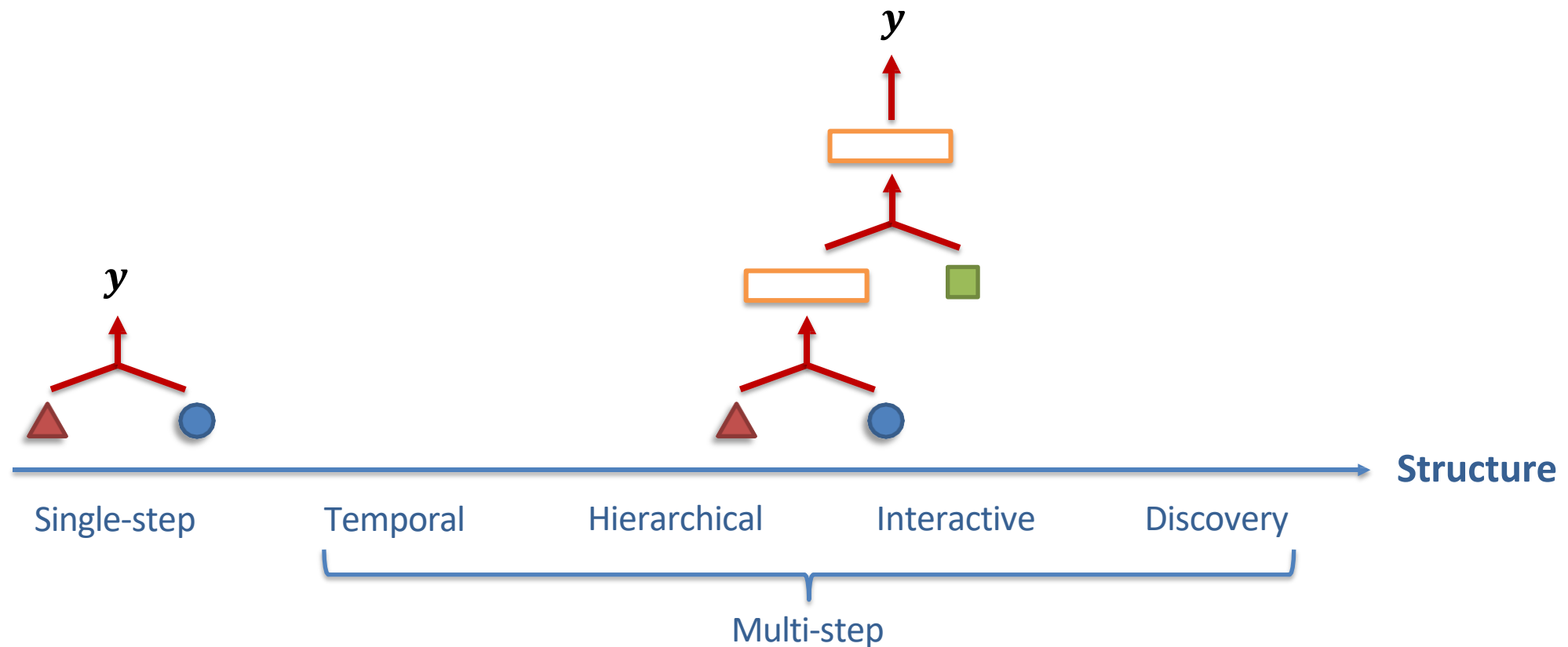
CLIP, ViLT, ViLBERT, etc.
Often near chance

Compositional Generalization
to novel combinations outside
of training data

1. Structure: <subject> <verb> <object>
2. Concepts: 'plants', 'lightbulb'
3. Inference: 'surrounding' – spatial relation
4. Knowledge: from humans!

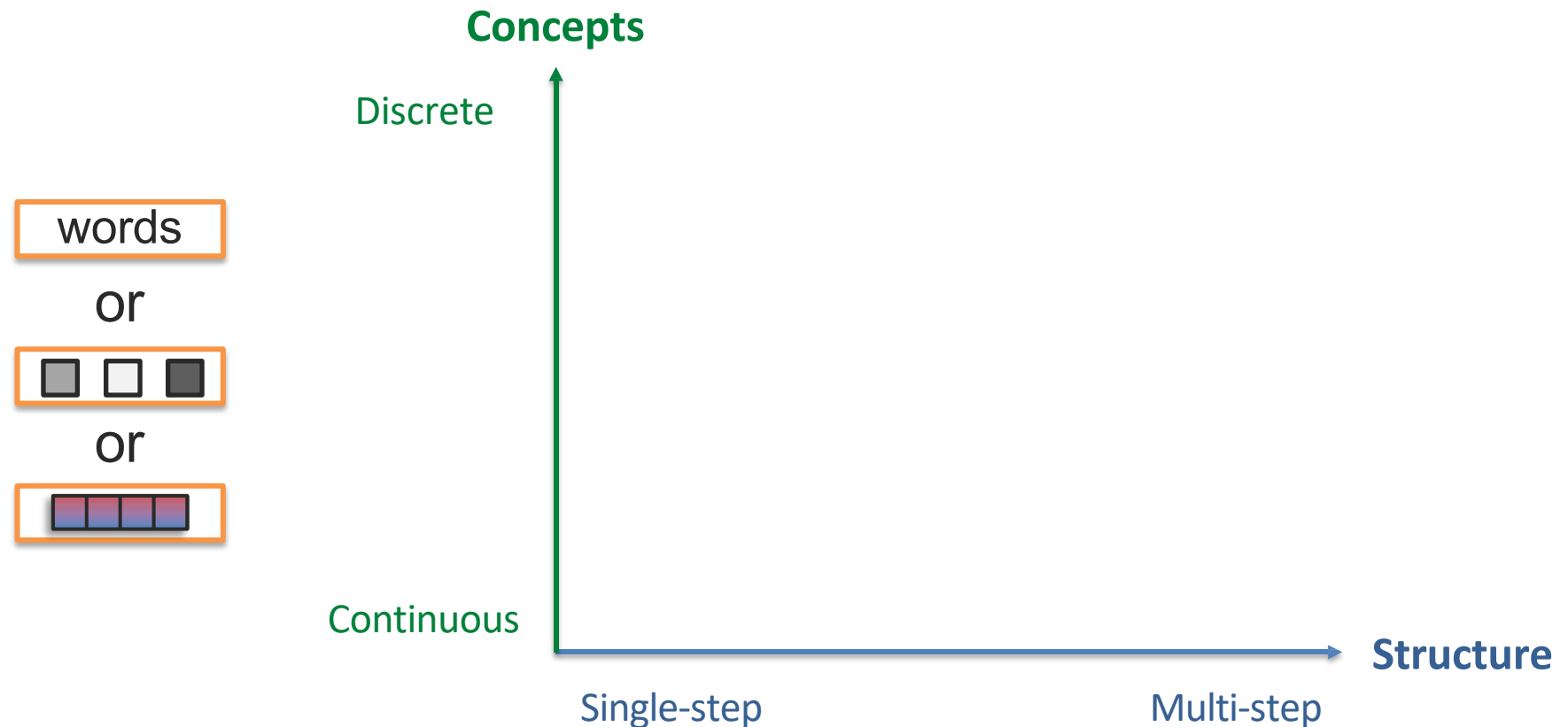
Sub-Challenge 3a: Structure Modeling

Definition: Defining or learning the relationships over which reasoning occurs.



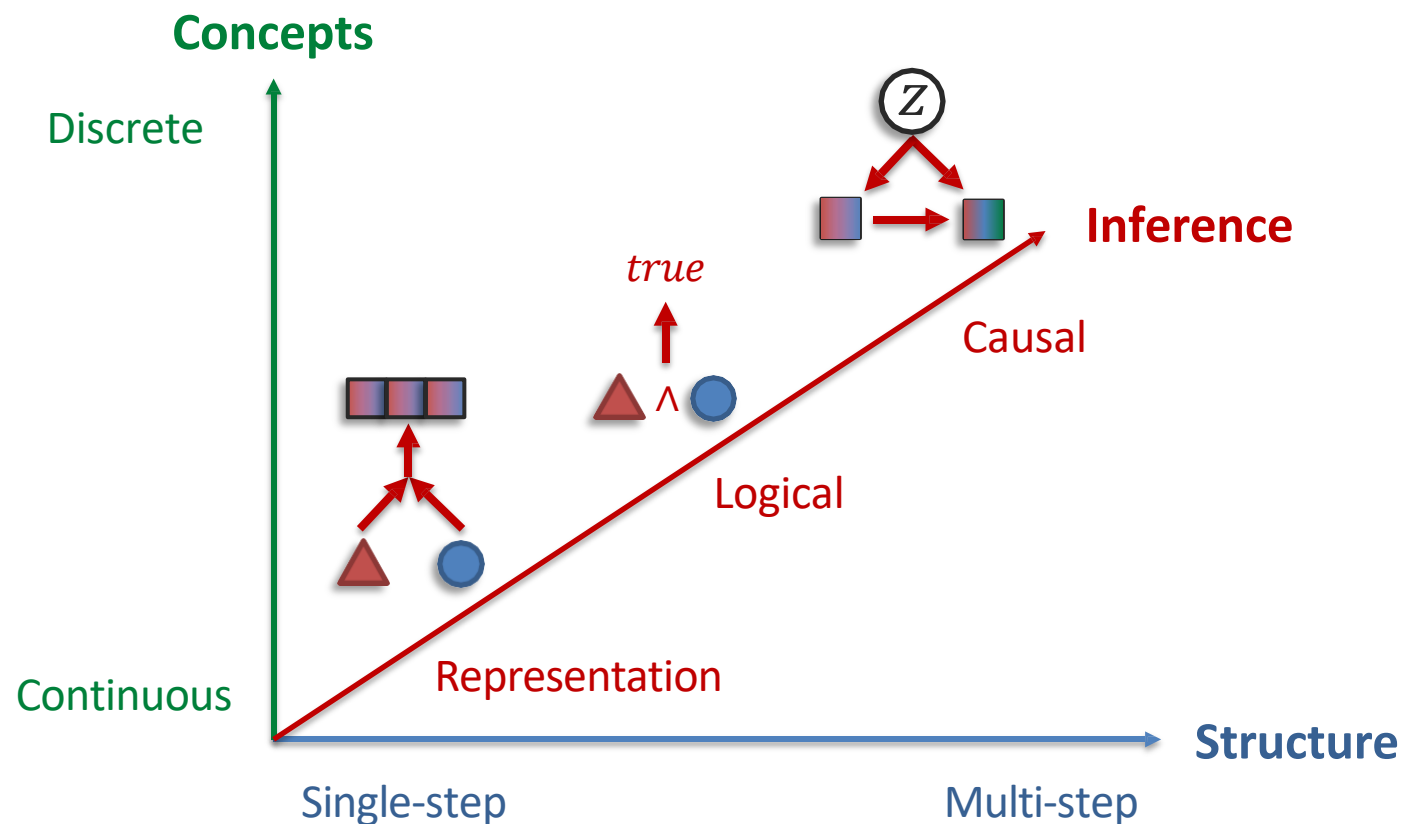
Sub-Challenge 3b: Intermediate Concepts

Definition: The parameterization of individual multimodal concepts in the reasoning process.



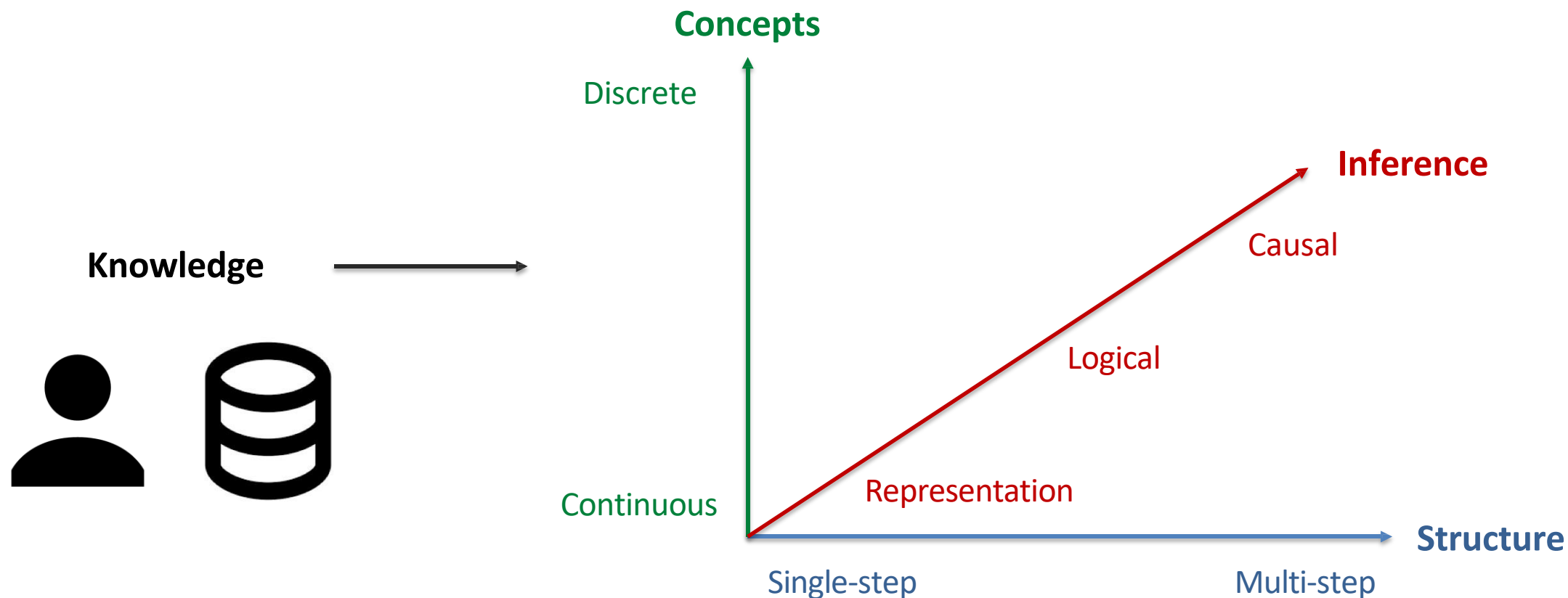
Sub-Challenge 3c: Inference Paradigm

Definition: How increasingly abstract concepts are inferred from individual multimodal evidences.



Sub-Challenge 3d: External Knowledge

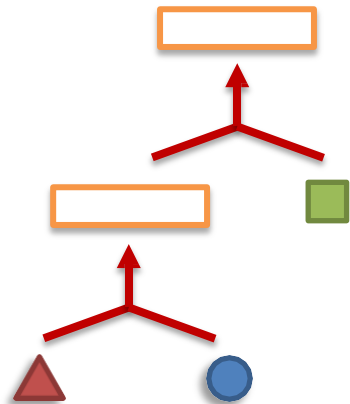
Definition: Leveraging external knowledge in the study of structure, concepts, and inference.



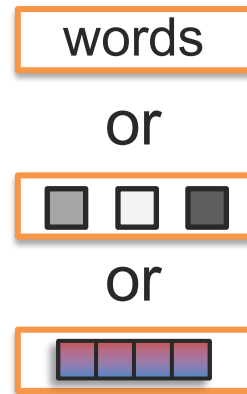
Summary: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure.

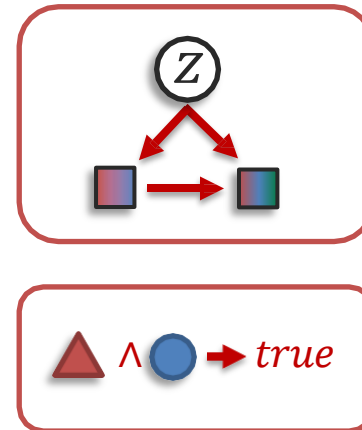
(A) Structure modeling



(B) Intermediate concepts



(C) Inference paradigm

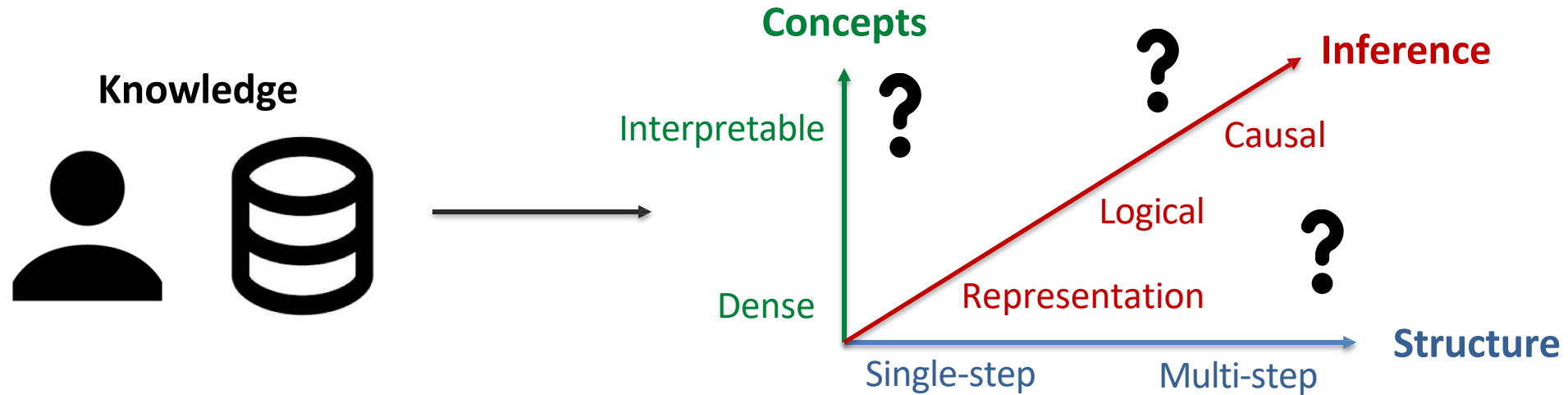


(D) External knowledge



More Reasoning

Open
challenges



Open challenges:

- Structure: multi-step inference
- Concepts: interpretable + differentiable representations
- Composition: explicit, logical, causal...
- Knowledge: integrating explicit knowledge with pretrained models
- Probing pretraining models for reasoning capabilities